

Vector Calculus

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(adapted from Fieller (2016), [Section 7.2](#))

This section covers derivatives of functions of vectors and matrices. Its linear algebra prerequisites, such as vectors, matrices, transposes, dot products, and quadratic forms, are covered in [Linear Algebra](#).

Let $\tilde{x}$ and $\tilde{\beta}$ be column vectors of length p (see [column vector](#) and [dot product](#)).

Definition 0.1 (Vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) \\ \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) \\ \vdots \\ \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Example 0.1 (A vector derivative). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ and $f(\tilde{\beta}) = 3\beta_1 + 5\beta_2^2$. Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} (3\beta_1 + 5\beta_2^2) \\ \frac{\partial}{\partial \beta_2} (3\beta_1 + 5\beta_2^2) \end{bmatrix} = \begin{bmatrix} 3 \\ 10\beta_2 \end{bmatrix},$$

and at $\tilde{\beta} = (1, 2)^\top$ the vector derivative is $(3, 20)^\top$.

Video lecture

Hutchinson's [Gradients Refresher](#) (17 min) covers gradients and partial derivatives, the ideas behind this section (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

💡 Videos

Jon Krohn’s “Calculus for Machine Learning” YouTube playlist has videos on partial derivatives:

- [What Partial Derivatives Are \(Hands-on Introduction\)](#)
- [Partial Derivative Exercises](#)
- [Advanced Partial Derivatives](#)
- [Advanced Partial-Derivative Exercises](#)
- [Partial Derivative Notation](#)

Checking a gradient by its shape

The gradient always has the same shape as the input it is taken with respect to, because it holds exactly one partial derivative per component of that input:

Table 1: The shape of a gradient follows the shape of what it differentiates with respect to.

f is a function of	its gradient is
a single number	a single number — the ordinary derivative
a vector in $\mathbb{R}^p$	a vector in $\mathbb{R}^p$
a matrix in $\mathbb{R}^{m \times n}$	a matrix in $\mathbb{R}^{m \times n}$

The table is the cheapest check there is on a gradient just worked out by hand: if the shapes disagree, something went wrong. It is the shape rule of matrix multiplication again, applied to calculus.

The course website’s version of this page has an interactive plot [here](#).

Exercise 0.1 (Take a gradient by hand). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be

$$f(\tilde{w}) = 3w_1w_2 + w_2^3 - 5w_1$$

1. Compute $\partial f / \partial w_1$, treating w_2 as a constant.
2. Compute $\partial f / \partial w_2$, treating w_1 as a constant.
3. Assemble $\nabla_{\tilde{w}} f$ and evaluate it at $\tilde{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$.
4. Check the shape of your answer against Table [1](#).

Solution

Solution 0.1. 1. Differentiate with respect to w_1 , holding w_2 fixed. The first term is the constant $3w_2$ times w_1 , so it contributes $3w_2$. The second term has no w_1 in it, so it contributes 0. The third term is -5 times w_1 , so it contributes -5 :

$$\frac{\partial f}{\partial w_1} = 3w_2 + 0 - 5 = 3w_2 - 5$$

2. Now with respect to w_2 , holding w_1 fixed. The first term is the constant $3w_1$ times w_2 , contributing $3w_1$. The second term contributes $3w_2^2$. The third has no w_2 in it, contributing 0:

$$\frac{\partial f}{\partial w_2} = 3w_1 + 3w_2^2 + 0 = 3w_1 + 3w_2^2$$

3. Stack the two, in the order the coordinates are numbered:

$$\nabla_{\tilde{w}} f(\tilde{w}) = \begin{bmatrix} 3w_2 - 5 \\ 3w_1 + 3w_2^2 \end{bmatrix}$$

The gradient is a vector-valued *function* of $\tilde{w}$, not a single vector. At $\tilde{w} = \begin{bmatrix} 2 & -1 \end{bmatrix}^\top$,

$$\nabla_{\tilde{w}} f = \begin{bmatrix} 3(-1) - 5 \\ 3(2) + 3(-1)^2 \end{bmatrix} = \begin{bmatrix} -3 - 5 \\ 6 + 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

4. The input was a vector in $\mathbb{R}^2$ and so is the answer, as Table 1 requires. Note the signs: from this point, increasing w_1 *decreases* f while increasing w_2 increases it, so the uphill direction is neither axis.

Definition 0.2 (Row-vector derivative). If $f(\tilde{\beta})$ is a scalar-valued function of a $p \times 1$ vector $\tilde{\beta}$, such as $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$, then its **row-vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) & \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) & \cdots & \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

Theorem 0.1 (Row and column derivatives are transposes).

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \left(\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) \right)^\top$$

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \left(\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) \right)^\top$$

i Proof

Proof. By Definition 0.1 and Definition 0.2, entry j of both $\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta})$ and $\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta})$ is $\frac{\partial}{\partial \beta_j} f(\tilde{\beta})$; the first is a $p \times 1$ column and the second a $1 \times p$ row with the same entries in the same order, so each is the transpose of the other. $\square$

Example 0.2 (Row and column derivatives of a linear function). For $f(\tilde{\beta}) = 3\beta_1 + 5\beta_2$:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = [3 \quad 5],$$

and each is the transpose of the other.

Definition 0.3 (Derivative of a vector-valued function). If $\tilde{y} = \tilde{y}(\tilde{\beta}) = (y_1, \dots, y_q)^\top$ is a $q \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, its **derivative with respect to $\tilde{\beta}$** is the $p \times q$ matrix whose (i, j) entry is

$$\left[\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top \right]_{ij} \stackrel{\text{def}}{=} \frac{\partial}{\partial \beta_i} y_j, \quad i = 1, \dots, p, \quad j = 1, \dots, q.$$

These notes use this *denominator layout* throughout: rows index the entries of $\tilde{\beta}$ (the denominator) and columns index the entries of $\tilde{y}$ (the numerator), so column j is the vector derivative $\frac{\partial}{\partial \tilde{\beta}} y_j$ (Definition 0.1). Both $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}$ and $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top$ denote this $p \times q$ matrix.

Example 0.3 (Differentiating a 3×1 function of a 2×1 vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ ($p = 2$) and $\tilde{y}(\tilde{\beta}) = (\beta_1^2, \beta_1\beta_2, 3\beta_2)^\top$ ($q = 3$). Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top}_{2 \times 3} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} \beta_1^2 & \frac{\partial}{\partial \beta_1} \beta_1\beta_2 & \frac{\partial}{\partial \beta_1} 3\beta_2 \\ \frac{\partial}{\partial \beta_2} \beta_1^2 & \frac{\partial}{\partial \beta_2} \beta_1\beta_2 & \frac{\partial}{\partial \beta_2} 3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & \beta_2 & 0 \\ 0 & \beta_1 & 3 \end{bmatrix}$$

Remark 0.1 (Numerator layout). Some sources use the transposed, *numerator layout*, in which the derivative is the $q \times p$ Jacobian matrix with (j, i) entry $\frac{\partial}{\partial \beta_i} y_j$. For example, in numerator layout, the derivative in Example 0.3 is the 3×2 matrix

$$\begin{bmatrix} 2\beta_1 & 0 \\ \beta_2 & \beta_1 \\ 0 & 3 \end{bmatrix}$$

the transpose of the 2×3 matrix found there. Check a source's layout before combining its formulas with these.

Definition 0.4 (Constant). A $q \times 1$ vector $\tilde{x}$ is **constant with respect to** the $p \times 1$ vector $\tilde{\beta}$ if its derivative (Definition 0.3) is zero:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{p \times q} = \underbrace{\mathbf{0}}_{p \times q}$$

Example 0.4 (A constant vector). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$ and $\tilde{x} = (3, 5)^\top$, so $x_1 = 3$ and $x_2 = 5$ do not depend on $\tilde{\beta}$. Expanding $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top$ into its matrix of scalar partial derivatives (Definition 0.3) and evaluating each entry:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{2 \times 2} = \frac{\partial}{\partial \tilde{\beta}} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} x_1 & \frac{\partial}{\partial \beta_1} x_2 \\ \frac{\partial}{\partial \beta_2} x_1 & \frac{\partial}{\partial \beta_2} x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} 3 & \frac{\partial}{\partial \beta_1} 5 \\ \frac{\partial}{\partial \beta_2} 3 & \frac{\partial}{\partial \beta_2} 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \underbrace{\mathbf{0}}_{2 \times 2}$$

Every entry is the derivative of a constant, so $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top = \underbrace{\mathbf{0}}_{2 \times 2}$ and $\tilde{x}$ is constant with respect to $\tilde{\beta}$ (Definition 0.4).

Example 0.5 (A vector that is not constant). With the same $\tilde{\beta}$, let $\tilde{x} = (\beta_1, 3)^\top$. Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{2 \times 2} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} \beta_1 & \frac{\partial}{\partial \beta_1} 3 \\ \frac{\partial}{\partial \beta_2} \beta_1 & \frac{\partial}{\partial \beta_2} 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

which is not $\underbrace{\mathbf{0}}_{2 \times 2}$, so this $\tilde{x}$ is not constant with respect to $\tilde{\beta}$: its first entry changes when β_1 does.

Theorem 0.2 (Derivative of a dot product). *If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{x} \cdot \tilde{\beta})}_{p \times 1} = \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta} \cdot \tilde{x})}_{p \times 1} = \underbrace{\tilde{x}}_{p \times 1}$$

i Proof

Proof.

$$\begin{aligned}\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) &= \begin{bmatrix} \frac{\partial}{\partial \beta_1}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \frac{\partial}{\partial \beta_2}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \vdots \\ \frac{\partial}{\partial \beta_p}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \\ &= \tilde{x}\end{aligned}$$

□

Example 0.6 (Derivative of a dot product). Let $\tilde{x} = (3, 5)^\top$ (constant with respect to $\tilde{\beta}$; see Example 0.4) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x} \cdot \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by Theorem 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta})}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Verifying entry-wise:

$$\frac{\partial}{\partial \tilde{\beta}}(3\beta_1 + 5\beta_2) = \begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1 + 5\beta_2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1 + 5\beta_2) \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Both methods agree.

Theorem 0.3 (Product rule for dot-products). *If $\tilde{a} = \tilde{a}(\tilde{x})$ and $\tilde{b} = \tilde{b}(\tilde{x})$ are differentiable $p \times 1$ vector functions of $\tilde{x}$, then:*

$$\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}}_{p \times 1} \cdot \underbrace{\tilde{b}}_{p \times 1} = \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}^\top}_{1 \times p} \right) \underbrace{\tilde{b}}_{p \times 1} + \left(\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{b}^\top}_{1 \times p} \right) \underbrace{\tilde{a}}_{p \times 1}$$

i Proof

Proof. Entry-wise, for $i = 1, \dots, p$:

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{x}} (\tilde{a} \cdot \tilde{b}) \right]_i &= \frac{\partial}{\partial x_i} \sum_{k=1}^p a_k b_k \\
&= \sum_{k=1}^p \left(b_k \frac{\partial}{\partial x_i} a_k + a_k \frac{\partial}{\partial x_i} b_k \right) \\
&= \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{a}^\top \right) \tilde{b} \right]_i + \left[\left(\frac{\partial}{\partial \tilde{x}} \tilde{b}^\top \right) \tilde{a} \right]_i
\end{aligned}$$

□

Example 0.7 (Example of the dot-product rule). Apply Theorem 0.3 with the vector $\tilde{\beta} = (\beta_1, \beta_2)^\top$ in the role of $\tilde{x}$. Let $\tilde{a}(\tilde{\beta}) = (\beta_1, \beta_1 \beta_2)^\top$ and $\tilde{b}(\tilde{\beta}) = (\beta_2, \beta_1)^\top$. Then:

$$\tilde{a} \cdot \tilde{b} = \beta_1 \cdot \beta_2 + \beta_1 \beta_2 \cdot \beta_1 = \beta_1 \beta_2 + \beta_1^2 \beta_2$$

By direct calculation:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} (\tilde{a} \cdot \tilde{b}) = \frac{\partial}{\partial \tilde{\beta}} (\beta_1 \beta_2 + \beta_1^2 \beta_2) = \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1 + \beta_1^2 \end{pmatrix}$$

By the product rule (Theorem 0.3), using $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{a}^\top}_{2 \times 2} = \begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}$ and $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{b}^\top}_{2 \times 2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{2 \times 1} (\tilde{a} \cdot \tilde{b}) &= \underbrace{\begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_2 \\ \beta_1 \end{pmatrix}}_{2 \times 1} + \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_1 \\ \beta_1 \beta_2 \end{pmatrix}}_{2 \times 1} \\
&= \begin{pmatrix} \beta_2 + \beta_1 \beta_2 \\ \beta_1^2 \end{pmatrix} + \begin{pmatrix} \beta_1 \beta_2 \\ \beta_1 \end{pmatrix} \\
&= \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1^2 + \beta_1 \end{pmatrix}
\end{aligned}$$

Both methods agree.

Theorem 0.4 (Derivative of a linear map). If $\mathbf{A}$ is an $m \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}}_{p \times m} (\mathbf{A} \tilde{\beta}) = \underbrace{\mathbf{A}^\top}_{p \times m}$$

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ (see Definition 0.3) and column j indexes the numerator $\mathbf{A}\tilde{\beta}$:

$$\begin{aligned} \left[\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{\beta}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A}\tilde{\beta})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^p a_{jk} \beta_k \\ &= a_{ji} \\ &= [\mathbf{A}^\top]_{ij} \end{aligned}$$

□

Example 0.8 (Derivative of a linear map). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ (1×2) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{A}\tilde{\beta} = 2\beta_1 + 3\beta_2$, and by Theorem 0.4:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{\beta})}_{2 \times 1} = \underbrace{\mathbf{A}^\top}_{2 \times 1} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Theorem 0.5 (Vector-derivative of a matrix-vector product). *If $\mathbf{A}$ is an $m \times q$ matrix that is constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{v})}_{p \times m} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{A}^\top}_{q \times m}$$

i Proof

Proof. For entry (i, j) , where row i indexes the denominator $\tilde{\beta}$ and column j indexes the numerator $\mathbf{A}\tilde{v}$ (see Definition 0.3):

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{v}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A} \tilde{v})_j \\
&= \frac{\partial}{\partial \beta_i} \sum_{k=1}^q a_{jk} v_k \\
&= \sum_{k=1}^q a_{jk} \frac{\partial}{\partial \beta_i} v_k \\
&= \sum_{k=1}^q \left[\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right]_{ik} [\mathbf{A}^\top]_{kj} \\
&= \left[\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{A}^\top \right]_{ij}
\end{aligned}$$

□

Example 0.9 (Vector-derivative of a matrix-vector product). Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ (1×2 , constant) and $\tilde{v}(\tilde{\beta}) = (\beta_1^2, \beta_2^2)^\top$. Then $\mathbf{A} \tilde{v} = 2\beta_1^2 + 3\beta_2^2$. By Theorem 0.5:

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{v})}_{2 \times 1} &= \begin{pmatrix} 2\beta_1 & 0 \\ 0 & 2\beta_2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\
&= \begin{pmatrix} 4\beta_1 \\ 6\beta_2 \end{pmatrix}
\end{aligned}$$

Remark 0.2 (The derivative of a linear map as a special case). This result generalizes Theorem 0.4, which is the special case $\tilde{v} = \tilde{\beta}$ (so that $q = p$, $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}_p$, and $\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \tilde{\beta}) = \mathbf{I}_p \mathbf{A}^\top = \mathbf{A}^\top$). For example, Example 0.8 is the case $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$ and $\tilde{v} = \tilde{\beta} = (\beta_1, \beta_2)^\top$, where this result gives $\mathbf{I}_2 \mathbf{A}^\top = (2, 3)^\top$.

Theorem 0.6 (Vector-derivative of a product of matrices). *If $\mathbf{A}$ ($\ell \times m$) and $\mathbf{B}$ ($m \times q$) are constant with respect to $\tilde{\beta}$, and $\tilde{v} = \tilde{v}(\tilde{\beta})$ is a $q \times 1$ vector that depends on the $p \times 1$ vector $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A} \mathbf{B} \tilde{v})}_{p \times \ell} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{B}^\top}_{q \times m} \underbrace{\mathbf{A}^\top}_{m \times \ell}$$

i Proof

Proof. Apply Theorem 0.5 with the constant $\ell \times q$ matrix $\mathbf{AB}$, then use the [transpose of a product](#):

$$\begin{aligned}\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) (\mathbf{AB})^\top \quad (\text{derivative of a matrix-vector product}) \\ &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{B}^\top \mathbf{A}^\top \quad (\text{transpose of a product})\end{aligned}$$

□

Example 0.10. Let $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ (1×2), $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2), and $\tilde{v}(\tilde{\beta}) = \tilde{\beta}$ where $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\mathbf{AB}\tilde{v} = 2\beta_1$, and:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v})}_{2 \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right)}_{2 \times 2} \underbrace{\mathbf{B}^\top \mathbf{A}^\top}_{2 \times 2 \times 1} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Corollary 0.1 (Derivative of a dot product, transpose-product form). *If $\tilde{x}$ is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta})}_{p \times 1} = \tilde{x}_{p \times 1}$$

i Proof

Proof. Using Theorem 0.2:

Since $\tilde{x}^\top \tilde{\beta} = \tilde{x} \cdot \tilde{\beta}$ (see [dot product](#)), and $\tilde{x}$ is constant with respect to $\tilde{\beta}$:

$$\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta}) = \frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) = \tilde{x}$$

by Theorem 0.2.

□

i Proof

Proof. Using **Theorem 0.5**:

Since $\tilde{x}$ is constant with respect to $\tilde{\beta}$, $\mathbf{A} = \tilde{x}^\top$ is a constant $1 \times p$ matrix. Applying **Theorem 0.5** with $\tilde{v} = \tilde{\beta}$ (so $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} (\tilde{x}^\top \tilde{\beta}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right) (\tilde{x}^\top)^\top \\ &= \mathbf{I} \cdot \tilde{x} \\ &= \tilde{x} \end{aligned}$$

□

Example 0.11 (Derivative of a transpose product). Let $\tilde{x} = (3, 5)^\top$ and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{x}^\top \tilde{\beta} = 3\beta_1 + 5\beta_2$, and by **Corollary 0.1**:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \left(\underbrace{\tilde{x}^\top}_{1 \times 2} \underbrace{\tilde{\beta}}_{2 \times 1} \right)}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Remark 0.3 (The coefficient gets transposed). This vector derivative formula looks a lot like non-vector calculus, except that you have to transpose the coefficient: in scalar calculus $\frac{\partial}{\partial x}(cx) = c$, but here the coefficient $\tilde{x}^\top$ (a row vector) becomes $\tilde{x}$ (a column vector) in the result. For example, with $\tilde{x} = (2, -1)^\top$, $\tilde{x}^\top \tilde{\beta} = 2\beta_1 - \beta_2$, whose vector derivative is the column vector $(2, -1)^\top = \tilde{x}$, not the row vector $\tilde{x}^\top = (2, -1)$.

Theorem 0.7 (Derivative of a quadratic form). *For a quadratic form (see [quadratic form](#)), if $\mathbf{S}$ is a symmetric $p \times p$ matrix that is constant with respect to $\tilde{\beta}$, then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{p \times 1} = \underbrace{2\mathbf{S} \tilde{\beta}}_{p \times 1}$$

i Proof

Proof. Expanding entry-wise, $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k$. Differentiating component-wise with respect to β_i for $i = 1, \dots, p$:

$$\begin{aligned}
\left[\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) \right]_i &= \frac{\partial}{\partial \beta_i} \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k \quad (\text{expand quadratic form}) \\
&= \sum_{k=1}^p s_{ik} \beta_k + \sum_{j=1}^p s_{ji} \beta_j \quad (\text{product rule for } \beta_i \beta_k) \\
&= [\mathbf{S} \tilde{\beta}]_i + [\mathbf{S}^\top \tilde{\beta}]_i \quad (\text{matrix-vector multiplication definition}) \\
&= [(\mathbf{S} + \mathbf{S}^\top) \tilde{\beta}]_i \quad (\text{linearity of matrix multiplication})
\end{aligned}$$

When $\mathbf{S}$ is symmetric ($\mathbf{S} = \mathbf{S}^\top$), $\mathbf{S} + \mathbf{S}^\top = 2\mathbf{S}$, so $\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S} \tilde{\beta}$. $\square$

Remark 0.4 (Like the derivative of cx^2). This operation is like taking the derivative of cx^2 with respect to x in non-vector calculus: $\frac{\partial}{\partial x}(cx^2) = 2cx$, and Theorem 0.7 says $\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S} \tilde{\beta}$. For example, with $p = 1$, $\mathbf{S} = (3)$, and $\tilde{\beta} = (\beta_1)$, $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2$, and its derivative is $6\beta_1 = 2\mathbf{S} \tilde{\beta}$.

Example 0.12 (Derivative of a quadratic form). Let $\mathbf{S} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ (2×2 , symmetric and constant) and $\tilde{\beta} = (\beta_1, \beta_2)^\top$. Then $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2$. By Theorem 0.7:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{2 \times 1} = 2\mathbf{S} \tilde{\beta} = 2 \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Differentiating component-wise directly:

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \\ \frac{\partial}{\partial \beta_2} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Both methods agree.

Corollary 0.2 (Derivative of a simple quadratic form).

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \tilde{\beta})}_{p \times 1} = \underbrace{2\tilde{\beta}}_{p \times 1}$$

Proof

Proof. Applying Theorem 0.7 with $\mathbf{S} = \mathbf{I}_{p \times p}$ (which is symmetric and constant with respect to $\tilde{\beta}$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta}) &= \frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{I}_{p \times p} \tilde{\beta}) \quad (\text{rewrite with identity matrix}) \\ &= 2\mathbf{I}_{p \times p} \tilde{\beta} \quad (\text{derivative of a quadratic form, with } \mathbf{S} = \mathbf{I}_{p \times p}) \\ &= 2\tilde{\beta} \quad (\text{identity matrix property}) \end{aligned}$$

□

Remark 0.5 (Like the derivative of x^2). This vector derivative is like taking the derivative of x^2 : in scalar calculus $\frac{\partial}{\partial x} x^2 = 2x$, and Corollary 0.2 says $\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta}) = 2\tilde{\beta}$. For example, with $p = 1$ and $\tilde{\beta} = (\beta_1)$, $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2$, and its derivative is $2\beta_1 = 2\tilde{\beta}$.

Example 0.13 (Derivative of a sum of squares). Let $\tilde{\beta} = (\beta_1, \beta_2)^\top$, so $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2 + \beta_2^2$. By Corollary 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta})}_{2 \times 1} = 2\tilde{\beta} = \begin{pmatrix} 2\beta_1 \\ 2\beta_2 \end{pmatrix}$$

Direct partial differentiation yields the same column vector.

Videos

Jon Krohn’s “Calculus for Machine Learning” YouTube playlist has videos on gradients of squared-error costs:

- [The Gradient of Quadratic Cost](#)
- [The Gradient of Mean Squared Error](#)

Theorem 0.8 (Vector chain rule). Let $\tilde{x}$ be a $p \times 1$ vector, let $\tilde{y} = \tilde{g}(\tilde{x})$ be a $q \times 1$ vector-valued function of $\tilde{x}$, and let $z = f(\tilde{y})$ be a scalar-valued function of $\tilde{y}$, where $\tilde{g}$ and f have continuous partial derivatives. Then $z = f(\tilde{g}(\tilde{x}))$, as a function of $\tilde{x}$, satisfies

$$\underbrace{\frac{\partial z}{\partial \tilde{x}}}_{p \times 1} = \underbrace{\frac{\partial \tilde{y}}{\partial \tilde{x}}}_{p \times q} \underbrace{\frac{\partial z}{\partial \tilde{y}}}_{q \times 1}$$

where $\frac{\partial \tilde{y}}{\partial \tilde{x}}$ is the derivative of Definition 0.3 and $\frac{\partial z}{\partial \tilde{x}}$ and $\frac{\partial z}{\partial \tilde{y}}$ are vector derivatives (Definition 0.1).

Remark 0.6 (The order of the factors matters). The vector chain rule (Theorem 0.8) is like the univariate [chain rule](#), but the order matters now: $\frac{\partial \tilde{y}}{\partial \tilde{x}}$ is $p \times q$ and $\frac{\partial z}{\partial \tilde{y}}$ is $q \times 1$, so the product $\frac{\partial z}{\partial \tilde{y}} \frac{\partial \tilde{y}}{\partial \tilde{x}}$ in the other order is not even defined unless $p = 1$. The version presented here is for the [gradient](#) (column vector); the [total derivative](#) (row vector) would be the [transpose of the gradient](#), $(\frac{\partial z}{\partial \tilde{x}})^\top = (\frac{\partial z}{\partial \tilde{y}})^\top (\frac{\partial \tilde{y}}{\partial \tilde{x}})^\top$, with the factors in the reverse order.

Example 0.14 (Applying the vector chain rule). Let $\tilde{x} = (x_1, x_2)^\top$, $\tilde{y} = \tilde{g}(\tilde{x}) = (x_1 + x_2, x_1 x_2)^\top$, and $z = f(\tilde{y}) = y_1^2 + y_2$. Then

$$\begin{aligned}
 \frac{\partial z}{\partial \tilde{x}} &= \frac{\partial \tilde{y}}{\partial \tilde{x}} \frac{\partial z}{\partial \tilde{y}} && \text{(vector chain rule)} \\
 \underset{2 \times 1}{\frac{\partial z}{\partial \tilde{x}}} &= \underset{2 \times 2}{\frac{\partial \tilde{y}}{\partial \tilde{x}}} \underset{2 \times 1}{\frac{\partial z}{\partial \tilde{y}}} && \text{(differentiate } \tilde{g} \text{ and } f) \\
 &= \begin{bmatrix} 1 & x_2 \\ 1 & x_1 \end{bmatrix} \begin{bmatrix} 2y_1 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 2(x_1 + x_2) + x_2 \\ 2(x_1 + x_2) + x_1 \end{bmatrix} && \text{(multiply, and substitute } y_1 = x_1 + x_2) \\
 &= \begin{bmatrix} 2x_1 + 3x_2 \\ 3x_1 + 2x_2 \end{bmatrix} && \text{(collect terms)}
 \end{aligned}$$

This matches differentiating z directly: $z = (x_1 + x_2)^2 + x_1 x_2 = x_1^2 + 3x_1 x_2 + x_2^2$, so $\frac{\partial}{\partial x_1} z = 2x_1 + 3x_2$ and $\frac{\partial}{\partial x_2} z = 3x_1 + 2x_2$. The product in the other order, a 2×1 matrix times a 2×2 matrix, is not defined. The total derivative is the transpose, the row vector $(2x_1 + 3x_2, 3x_1 + 2x_2)$.

See <https://quickfem.com/finite-element-analysis/>, specifically https://quickfem.com/wp-content/uploads/IFEM.AppF_.pdf

See also https://en.wikipedia.org/wiki/Gradient#Relationship_with_Fr%C3%A9chet_derivative

Corollary 0.3 (Vector chain rule for quadratic forms). *If $\tilde{\varepsilon} = \tilde{\varepsilon}(\tilde{\beta})$ is an $n \times 1$ vector-valued function of the $p \times 1$ vector $\tilde{\beta}$, with continuous partial derivatives, then*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon}(\tilde{\beta}) \cdot \tilde{\varepsilon}(\tilde{\beta}))}_{p \times 1} = \underbrace{\left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon}(\tilde{\beta}) \right)}_{p \times n} \underbrace{(2\tilde{\varepsilon}(\tilde{\beta}))}_{n \times 1}$$

Proof

Proof. Apply Theorem 0.8 with $\tilde{x} = \tilde{\beta}$, $\tilde{y} = \tilde{\varepsilon}$, $q = n$, and $z = f(\tilde{\varepsilon}) = \tilde{\varepsilon} \cdot \tilde{\varepsilon} = \tilde{\varepsilon}^\top \tilde{\varepsilon}$:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) \frac{\partial}{\partial \tilde{\varepsilon}}(\tilde{\varepsilon}^\top \tilde{\varepsilon}) \quad (\text{vector chain rule}) \\ &= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) (2\tilde{\varepsilon}) \quad (\text{derivative of a simple quadratic form, in } \tilde{\varepsilon}) \end{aligned}$$

The second step is Corollary 0.2, applied to the $n \times 1$ vector $\tilde{\varepsilon}$ in place of $\tilde{\beta}$. □

Example 0.15 (Derivative of the residual sum of squares). Let $\tilde{y}$ ($n \times 1$) and $\mathbf{X}$ ($n \times p$) be constant with respect to $\tilde{\beta}$, and let $\tilde{\varepsilon}(\tilde{\beta}) = \tilde{y} - \mathbf{X}\tilde{\beta}$ be the vector of residuals. By Theorem 0.4, $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{X}\tilde{\beta}) = \mathbf{X}^\top$, and $\frac{\partial}{\partial \tilde{\beta}}\tilde{y} = \mathbf{0}_{p \times n}$ because $\tilde{y}$ is constant, so $\frac{\partial}{\partial \tilde{\beta}}\tilde{\varepsilon} = -\mathbf{X}^\top$ ($p \times n$). By Corollary 0.3:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= (-\mathbf{X}^\top)(2\tilde{\varepsilon}) \quad (\text{vector chain rule for quadratic forms}) \\ &= -2\mathbf{X}^\top(\tilde{y} - \mathbf{X}\tilde{\beta}) \quad (\text{substitute } \tilde{\varepsilon} = \tilde{y} - \mathbf{X}\tilde{\beta}) \end{aligned}$$

Setting this $p \times 1$ vector to $\tilde{0}$ gives the normal equations $\mathbf{X}^\top \mathbf{X}\tilde{\beta} = \mathbf{X}^\top \tilde{y}$ of least squares.

Videos

Jon Krohn’s “Calculus for Machine Learning” YouTube playlist has videos on the chain rule for partial derivatives:

- [The Chain Rule for Partial Derivatives](#)
- [Exercises on the Multivariate Chain Rule](#)

Definition 0.5 (Matrix derivative). For a scalar-valued function $f(\mathbf{X})$ of an $m \times n$ matrix $\mathbf{X}$, the **matrix derivative** is the $m \times n$ matrix whose (i, j) entry is the partial derivative of f with respect to the (i, j) entry of $\mathbf{X}$:

$$\left[\frac{\partial}{\partial \mathbf{X}} f \right]_{ij} = \frac{\partial}{\partial X_{ij}} f$$

Example 0.16 (The matrix derivative of a trace). Let $\mathbf{X}$ be a 2×2 matrix and $f(\mathbf{X}) = \text{tr}(\mathbf{X}) = X_{11} + X_{22}$ (see [trace](#)). Then $\frac{\partial}{\partial X_{ij}} f = 1$ if $i = j$ and 0 otherwise, so:

$$\frac{\partial}{\partial \mathbf{X}} f = \mathbf{I}_2$$

Theorem 0.9 (Matrix derivative of the trace of a matrix product). *If $\mathbf{A}$ ($r \times m$) and $\mathbf{B}$ ($n \times r$) are constant with respect to the $m \times n$ matrix $\mathbf{X}$, then:*

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{m \times n} = \underbrace{\mathbf{A}^\top}_{m \times r} \underbrace{\mathbf{B}^\top}_{r \times n}$$

i Proof

Proof. Write A_{kl} , X_{kl} , and B_{kl} for the entries of $\mathbf{A}$, $\mathbf{X}$, and $\mathbf{B}$. For entry (i, j) :

$$\begin{aligned} \left[\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB}) \right]_{ij} &= \frac{\partial}{\partial X_{ij}} \sum_{a=1}^r \sum_{b=1}^m \sum_{c=1}^n A_{ab} X_{bc} B_{ca} \quad (\text{trace of the } r \times r \text{ product } \mathbf{AXB}) \\ &= \sum_{a=1}^r A_{ai} B_{ja} \quad (\text{only the terms with } b = i, c = j \text{ depend on } X_{ij}) \\ &= \sum_{a=1}^r [\mathbf{A}^\top]_{ia} [\mathbf{B}^\top]_{aj} \quad (\text{definition of the transpose}) \\ &= [\mathbf{A}^\top \mathbf{B}^\top]_{ij} \quad (\text{definition of matrix multiplication}) \end{aligned}$$

□

Remark 0.7 (Why the theorem uses the trace). The trace makes $\text{tr}(\mathbf{AXB})$ a scalar, so its matrix derivative (Definition 0.5) is again an $m \times n$ matrix. The matrix product $\mathbf{AXB}$ itself (without the trace) is an $r \times r$ matrix, and each of its r^2 entries has a partial derivative with respect to each of the mn entries of $\mathbf{X}$. Those $r^2 mn$ partial derivatives form a four-index array (a fourth-order tensor), not a matrix, which is why this result is stated for the scalar $\text{tr}(\mathbf{AXB})$.

For example, with $\mathbf{A} = \mathbf{B} = \mathbf{I}_2$, the product $\mathbf{AXB} = \mathbf{X}$ has 4 entries, each with 4 partial derivatives, 16 in all, while its trace $X_{11} + X_{22}$ has the 4 partial derivatives that form the

2×2 matrix $\mathbf{I}_2$ of Example 0.16.

Example 0.17 (Differentiating a weighted trace). Let $\mathbf{A} = \mathbf{I}_2$ (2×2) and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (2×2). Then $\text{tr}(\mathbf{AXB}) = 2X_{11} + 3X_{22}$, and:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{2 \times 2} = \underbrace{\mathbf{A}^\top}_{2 \times 2} \underbrace{\mathbf{B}^\top}_{2 \times 2} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Exercise 0.2 (A gradient that does not mention its variable). Fix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix}$$

let $W \in \mathbb{R}^{2 \times 3}$ vary, and define

$$g(W) = \sum_{i=1}^2 \sum_{j=1}^3 W_{ij} A_{ij}$$

which is the entry-by-entry version of the inner product for matrices. Compute $\nabla_W g(W)$, and say what is unusual about the answer.

Solution

Solution 0.2. Written out over the six positions,

$$g(W) = 2W_{11} - W_{12} + 0W_{13} + 4W_{21} + 3W_{22} - 2W_{23}$$

Each W_{ij} appears in exactly one term, multiplied by A_{ij} and by nothing else, so differentiating with respect to it leaves A_{ij} behind:

$$\frac{\partial g}{\partial W_{ij}} = A_{ij}$$

Collecting those partial derivatives into a matrix of the same shape as W , as Table 1 requires,

$$\nabla_W g(W) = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix} = A$$

What is unusual is that the gradient came out **constant**: it does not mention W at all. The constant answer is not a quirk of this particular A . g is a linear function of W , and

the gradient of a linear function is constant everywhere, for the same reason the derivative of $f(w) = cw$ is c no matter where it is evaluated.

A constant gradient is the easy case, and it is not the case we usually face. Most objectives are curved — the squared error of a linear model is the standard example — so their gradient changes from point to point and the downhill direction has to be worked out afresh at every step.

Second derivatives and optimality conditions

Source

This section is adapted from the multivariate calculus and optimality-condition parts of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source states the second-order Taylor approximation and the optimality conditions without proof; here they are proved from the one-variable Taylor theorem, and the approximation is made exact by evaluating the Hessian at a point between $\tilde{z}$ and $\tilde{z} + \tilde{h}$. The source calls points with zero gradient critical points; here they are stationary points, because the [calculus notes](#) give “critical point” a wider meaning. These parts of the source are not part of this section:

- its one-variable Taylor example and its plots
- matrix calculus
- convexity
- Lagrange multipliers
- Newton’s method and gradient descent

A minimizer of a function of one variable has a flat tangent line ([flat point](#)), and, at a flat point, a positive second derivative guarantees a strict local minimum. This section extends both facts to a function f of a $p \times 1$ vector $\tilde{x}$: the gradient $\frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ (Definition 0.1) takes the place of the first derivative, and a $p \times p$ matrix of second partial derivatives takes the place of the second; both facts are proved in this section, for every $p \geq 1$. Throughout, $\frac{\partial}{\partial \tilde{x}} f(\tilde{z})$ means the gradient evaluated at $\tilde{x} = \tilde{z}$.

Definition 0.6 (Hessian matrix). Let f be a scalar-valued function of a $p \times 1$ vector $\tilde{x}$ whose first partial derivatives exist on an open ball around $\tilde{x}$ and whose second partial derivatives exist at $\tilde{x}$. The **Hessian matrix** of f at $\tilde{x}$ is the derivative (Definition 0.3) of the gradient $\frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ (Definition 0.1):

$$\underbrace{\mathbf{H}_f(\tilde{x})}_{p \times p} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) \right)^\top.$$

By Definition 0.3, with $y_j = \frac{\partial}{\partial x_j} f(\tilde{x})$, its (i, j) entry is

$$[\mathbf{H}_f(\tilde{x})]_{ij} = \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{x}) \right).$$

Example 0.18 (A Hessian matrix). Let $f(\tilde{x}) = e^{2x_1+x_2} - x_1$ for $\tilde{x} = (x_1, x_2)^\top$, and write $u = 2x_1 + x_2$. By the [chain rule](#), $\frac{\partial}{\partial x_1} e^u = 2e^u$ and $\frac{\partial}{\partial x_2} e^u = e^u$, so

$$\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) = \begin{bmatrix} 2e^u - 1 \\ e^u \end{bmatrix}.$$

Differentiating each entry again,

$$\mathbf{H}_f(\tilde{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1}(2e^u - 1) & \frac{\partial}{\partial x_1} e^u \\ \frac{\partial}{\partial x_2}(2e^u - 1) & \frac{\partial}{\partial x_2} e^u \end{bmatrix} = \begin{bmatrix} 4e^u & 2e^u \\ 2e^u & e^u \end{bmatrix},$$

and at $\tilde{x} = \tilde{0}$, where $u = 0$, $\mathbf{H}_f(\tilde{0}) = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$.

Example 0.19 (A function with a gradient but no Hessian at a point). Let $f(\tilde{x}) = x_1|x_1|$ for $\tilde{x} = (x_1, x_2)^\top$. For $x_1 > 0$, $f = x_1^2$ and $\frac{\partial}{\partial x_1} f = 2x_1$; for $x_1 < 0$, $f = -x_1^2$ and $\frac{\partial}{\partial x_1} f = -2x_1$; and at $x_1 = 0$ the difference quotient is $h|h|/h = |h| \rightarrow 0$. So the gradient exists everywhere: it is $(2|x_1|, 0)^\top$. But $2|x_1|$ has no derivative in x_1 at $x_1 = 0$ (its difference quotient $2|h|/h$ is 2 for $h > 0$ and -2 for $h < 0$), so $[\mathbf{H}_f(\tilde{x})]_{11}$, and with it the Hessian, does not exist at any $\tilde{x}$ with $x_1 = 0$.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on second and higher partial derivatives:

- [Higher-Order Partial Derivatives](#)
- [Exercise on Higher-Order Partial Derivatives](#)

Definition 0.7 (Continuity in several variables). A function f from $\mathbb{R}^p$ to $\mathbb{R}^q$ is **continuous at** $\tilde{x}$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\|f(\tilde{y}) - f(\tilde{x})\| < \varepsilon$ whenever $\|\tilde{y} - \tilde{x}\| < \delta$. It is **continuous on** a set if it is continuous at every point of the set. For $q = 1$, $\|f(\tilde{y}) - f(\tilde{x})\| = |f(\tilde{y}) - f(\tilde{x})|$. For $p = q = 1$, this definition is the usual [continuity](#), written out with the ε - δ definition of the limit $\lim_{y \rightarrow x} f(y) = f(x)$. Continuity survives the usual operations:

- A composition of continuous functions is continuous: choose the δ for the outer function first, and use it as the ε for the inner one.
- A sum $f + g$ of continuous real-valued functions is continuous, since $|(f + g)(\tilde{y}) - (f + g)(\tilde{x})| \leq |f(\tilde{y}) - f(\tilde{x})| + |g(\tilde{y}) - g(\tilde{x})|$: use $\varepsilon/2$ for each.
- A constant multiple cf is continuous, since $|cf(\tilde{y}) - cf(\tilde{x})| = |c| |f(\tilde{y}) - f(\tilde{x})|$: use $\varepsilon/(|c| + 1)$ for f .

Example 0.20 (A continuous function, and a discontinuous one).

- $f(\tilde{x}) = x_1 + x_2$ is continuous at every $\tilde{x}$. Each $|y_i - x_i| \leq \|\tilde{y} - \tilde{x}\|$, since the squared length is a sum of nonnegative squares, so

$$\begin{aligned} |f(\tilde{y}) - f(\tilde{x})| &= |(y_1 - x_1) + (y_2 - x_2)| && \text{(subtract)} \\ &\leq |y_1 - x_1| + |y_2 - x_2| && \text{(triangle inequality for numbers)} \\ &\leq 2 \|\tilde{y} - \tilde{x}\|, && \text{(each term is at most } \|\tilde{y} - \tilde{x}\|) \end{aligned}$$

which is less than ε whenever $\|\tilde{y} - \tilde{x}\| < \delta = \varepsilon/2$.

- $f(\tilde{x}) = 1$ if $x_1 > 0$ and $f(\tilde{x}) = 0$ otherwise is not continuous at $\tilde{0}$: for $\varepsilon = \frac{1}{2}$ and any $\delta > 0$, the point $\tilde{y} = (\delta/2, 0)^\top$ has $\|\tilde{y} - \tilde{0}\| = \delta/2 < \delta$ but $|f(\tilde{y}) - f(\tilde{0})| = 1 > \frac{1}{2}$.

Theorem 0.10 (Symmetry of the Hessian). *If the second partial derivatives of f exist and are continuous (Definition 0.7) on an open ball $\{\tilde{y} : \|\tilde{y} - \tilde{x}\| < r\}$ around $\tilde{x}$, for some $r > 0$, then $\frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{x}) \right) = \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_i} f(\tilde{x}) \right)$ for all i, j , so $\mathbf{H}_f(\tilde{x})$ (Definition 0.6) is symmetric (symmetric matrix).*

The proof applies the one-variable mean value theorem twice, which these notes do not develop; see (Rudin 1976, Theorem 9.41), which is stated for two variables: apply it to f as a function of x_i and x_j , with the other coordinates held fixed.

Example 0.21 (Mixed partial derivatives agree). In Example 0.18, the $(1, 2)$ and $(2, 1)$ entries of $\mathbf{H}_f(\tilde{x})$ are both $2e^{2x_1+x_2}$.

Example 0.22 (Without continuity, the mixed partials can differ). Let $f(x_1, x_2) = \frac{x_1 x_2 (x_1^2 - x_2^2)}{x_1^2 + x_2^2}$ for $\tilde{x} \neq \tilde{0}$, and $f(\tilde{0}) = 0$. For $x_2 \neq 0$, the derivative in x_1 at $(0, x_2)$ is

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(0, x_2) &= \lim_{h \rightarrow 0} \frac{f(h, x_2) - f(0, x_2)}{h} \quad (\text{definition of the partial derivative}) \\
&= \lim_{h \rightarrow 0} \frac{x_2(h^2 - x_2^2)}{h^2 + x_2^2} \quad (f(0, x_2) = 0; \text{cancel } h) \\
&= \frac{x_2 \cdot (-x_2^2)}{x_2^2} = -x_2, \quad (\text{the quotient is continuous at } h = 0)
\end{aligned}$$

and $\frac{\partial}{\partial x_1} f(\tilde{0}) = \lim_{h \rightarrow 0} (0 - 0)/h = 0$, so $\frac{\partial}{\partial x_1} f(0, x_2) = -x_2$ holds at $x_2 = 0$ too. In the same way, with the roles of x_1 and x_2 swapped, $f(x_1, k)/k = x_1(x_1^2 - k^2)/(x_1^2 + k^2) \rightarrow x_1$, so $\frac{\partial}{\partial x_2} f(x_1, 0) = x_1$ for every x_1 . So at $\tilde{0}$ (Definition 0.6)

$$[\mathbf{H}_f(\tilde{0})]_{21} = \frac{\partial}{\partial x_2} \left(\frac{\partial}{\partial x_1} f \right) = \frac{d}{dx_2} (-x_2) = -1, \quad [\mathbf{H}_f(\tilde{0})]_{12} = \frac{\partial}{\partial x_1} \left(\frac{\partial}{\partial x_2} f \right) = \frac{d}{dx_1} x_1 = 1 :$$

the Hessian at $\tilde{0}$ is not symmetric. By Theorem 0.10, then, the second partial derivatives of this f cannot all be continuous near $\tilde{0}$.

Theorem 0.11 (Hessian of a quadratic form). *If $\mathbf{S}$ is a symmetric $p \times p$ matrix that is constant with respect to $\tilde{x}$, then $f(\tilde{x}) = \tilde{x}^\top \mathbf{S} \tilde{x}$ has $\mathbf{H}_f(\tilde{x}) = 2\mathbf{S}$ for every $\tilde{x}$.*

Proof

Proof.

$$\begin{aligned}
\mathbf{H}_f(\tilde{x}) &= \frac{\partial}{\partial \tilde{x}} \left(\frac{\partial}{\partial \tilde{x}} (\tilde{x}^\top \mathbf{S} \tilde{x}) \right)^\top \quad (\text{Definition 6}) \\
&= \frac{\partial}{\partial \tilde{x}} (2\mathbf{S} \tilde{x})^\top \quad (\text{Theorem 7}) \\
&= \left(\frac{\partial}{\partial \tilde{x}} \tilde{x} \right) (2\mathbf{S})^\top \quad (\text{Theorem 5, with } \mathbf{A} = 2\mathbf{S} \text{ and } \tilde{v} = \tilde{x}; \frac{\partial}{\partial \tilde{x}} (2\mathbf{S} \tilde{x})^\top = \frac{\partial}{\partial \tilde{x}} (2\mathbf{S} \tilde{x}), \text{ Definition 3}) \\
&= \mathbf{I}_p (2\mathbf{S})^\top \quad (\frac{\partial}{\partial \tilde{x}} \tilde{x} = \mathbf{I}_p, \text{ Remark 2}) \\
&= 2\mathbf{S}. \quad (\mathbf{S} \text{ is symmetric})
\end{aligned}$$

□

Example 0.23 (The Hessian of a 2×2 quadratic form). For $\mathbf{S} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ as in Exam-

ple 0.12, Theorem 0.11 gives $\mathbf{H}_f(\tilde{x}) = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$. Directly, the gradient found there, with β_i renamed x_i , is $(6x_1 + 2x_2, 2x_1 + 4x_2)^\top$, and differentiating its entries by x_1 and by x_2 (Definition 0.6) gives

$$\mathbf{H}_f(\tilde{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1}(6x_1 + 2x_2) & \frac{\partial}{\partial x_1}(2x_1 + 4x_2) \\ \frac{\partial}{\partial x_2}(6x_1 + 2x_2) & \frac{\partial}{\partial x_2}(2x_1 + 4x_2) \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}.$$

Theorem 0.12 (A minimizer has derivative zero). *Let g be a real-valued function on an open interval containing c , and let g be differentiable at c . If c is a local minimizer of g , then $g'(c) = 0$.*

i Proof

Proof. Take $\delta > 0$ as in the definition of a local minimizer, shrunk if needed so that $(c - \delta, c + \delta)$ lies inside the interval; then $g(c) \leq g(c + h)$ whenever $|h| < \delta$. For $0 < h < \delta$, the difference quotient is

$$\frac{g(c + h) - g(c)}{h} \geq 0,$$

because its numerator is at least 0 and its denominator is positive. As $h \rightarrow 0$ from the right, these quotients tend to $g'(c)$ (the two-sided limit exists, so the one-sided limit equals it), and a limit of numbers that are all at least 0 is at least 0; so $g'(c) \geq 0$. For $-\delta < h < 0$, the numerator is still at least 0 but the denominator is negative, so the quotient is at most 0, and in the same way, with $h \rightarrow 0$ from the left, $g'(c) \leq 0$. Together, $g'(c) = 0$. $\square$

Example 0.24 (Applying the test, and its limits).

- $g(x) = (x - 2)^2$ has its minimizer at $c = 2$, and $g'(2) = 2(2 - 2) = 0$.
- The converse fails: $g(x) = x^3$ has $g'(0) = 3 \cdot 0^2 = 0$, but 0 is not a local minimizer, since $g(-h) = -h^3 < 0 = g(0)$ for every $h > 0$.
- Differentiability is needed: $g(x) = |x|$ has its minimizer at 0, but the difference quotient $|h|/h$ is 1 for $h > 0$ and -1 for $h < 0$, so it has no limit and $g'(0)$ does not exist.

Definition 0.8 (Stationary point). Let f be a scalar-valued function of a $p \times 1$ vector $\tilde{x}$. A point $\tilde{z}$ at which the gradient exists and

$$\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) = \tilde{0}_{p \times 1}$$

is a **stationary point** of f . For $p = 1$, this is a **flat point**.

Example 0.25 (A stationary point, and a point that is not one). Let $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$. Its gradient is $(2x_1 - 2, 2x_2)^\top$, which is $\tilde{0}_{2 \times 1}$ exactly when $x_1 = 1$ and $x_2 = 0$. So $(1, 0)^\top$ is its only stationary point, and $\tilde{0}$ is not one: the gradient there is $(-2, 0)^\top$.

Theorem 0.13 (First-order necessary condition). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. If $\tilde{x}^*$ is a **local minimizer** of f and the gradient of f exists at $\tilde{x}^*$, then $\tilde{x}^*$ is a stationary point of f (Definition 0.8).*

i Proof

Proof. Take $\delta > 0$ as in the definition of a local minimizer. Fix $i \in \{1, \dots, p\}$, let $\tilde{e}_i$ be the vector with 1 in entry i and 0 elsewhere, and let $g_i(t) \stackrel{\text{def}}{=} f(\tilde{x}^* + t\tilde{e}_i)$ for $t \in \mathbb{R}$. For $|t| < \delta$, $\|(\tilde{x}^* + t\tilde{e}_i) - \tilde{x}^*\| = |t| \|\tilde{e}_i\| = |t| < \delta$, so $g_i(0) = f(\tilde{x}^*) \leq f(\tilde{x}^* + t\tilde{e}_i) = g_i(t)$: 0 is a local minimizer of g_i . Moving $\tilde{x}$ from $\tilde{x}^*$ along $\tilde{e}_i$ changes only x_i , so $g'_i(0)$ is the partial derivative $\frac{\partial}{\partial x_i} f(\tilde{x}^*)$, which exists because the gradient does. By Theorem 0.12, $\frac{\partial}{\partial x_i} f(\tilde{x}^*) = g'_i(0) = 0$. The equation $g'_i(0) = 0$ holds for every i , so every entry of the gradient at $\tilde{x}^*$ is 0 (Definition 0.1). $\square$

Example 0.26 (Using the condition to locate, and to rule out, minimizers).

- For $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ of Example 0.25, any local minimizer must be the stationary point $(1, 0)^\top$. It is one: completing the square, $f(\tilde{x}) = (x_1 - 1)^2 + x_2^2 - 1 \geq -1 = f(1, 0)$.
- For $f(\tilde{x}) = e^{2x_1 + x_2} - x_1$ of Example 0.18, the second entry of the gradient is $e^{2x_1 + x_2} > 0$, so f has no stationary point, and so no local minimizer.
- The converse fails: $f(\tilde{x}) = x_1^2 - x_2^2$ has gradient $(2x_1, -2x_2)^\top$, so $\tilde{0}$ is a stationary point, but $f(0, t) = -t^2 < 0 = f(0)$ for every $t \neq 0$, so $\tilde{0}$ is not a local minimizer.

Theorem 0.14 (Taylor's theorem with a second-order remainder). *Let $a < b$, and let g be a real-valued function on $[a, b]$ whose derivative g' is **continuous** on $[a, b]$ and whose second derivative g'' exists at every point of (a, b) . Then there is a $\tau \in (a, b)$ with*

$$g(b) = g(a) + g'(a)(b - a) + \frac{1}{2} g''(\tau)(b - a)^2.$$

This theorem is the case $n = 2$ of (Rudin 1976, Theorem 5.15); its proof uses the mean value theorem, which these notes do not develop.

Example 0.27 (The remainder point for e^x). Take $g(x) = e^x$ on $[0, 1]$, so $g' = g'' = g$. Theorem 0.14 says $e = 1 + 1 + \frac{1}{2}e^\tau$ for some $\tau \in (0, 1)$. Solving, $e^\tau = 2(e - 2) \approx 2 \times 0.71828 = 1.43656$, so $\tau = \log 1.43656 \approx 0.362$, which is in $(0, 1)$.

Theorem 0.15 (Second-order Taylor theorem in several variables). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ (Definition 0.7), and let $\tilde{z}, \tilde{h} \in \mathbb{R}^p$. Then there is a $\tau \in (0, 1)$ with*

$$f(\tilde{z} + \tilde{h}) = f(\tilde{z}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) \right)^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau \tilde{h}) \tilde{h}.$$

i Proof

Proof. Let $g(t) \stackrel{\text{def}}{=} f(\tilde{z} + t\tilde{h})$ for $t \in \mathbb{R}$, the values of f along the line through $\tilde{z}$ in the direction $\tilde{h}$. The inner function $\tilde{y}(t) = \tilde{z} + t\tilde{h}$ has entries $y_j = z_j + th_j$, so $\frac{\partial}{\partial t} y_j = h_j$, and its derivative (Definition 0.3) is the $1 \times p$ matrix $\tilde{h}^\top$. By the vector chain rule (Theorem 0.8, with its input t of length 1, its inner function $\tilde{y}(t) = \tilde{z} + t\tilde{h}$, and its outer function f),

$$\begin{aligned} g'(t) &= \tilde{h}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{z} + t\tilde{h}) && \text{(Theorem 8)} \\ &= \sum_{j=1}^p h_j \frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}). && \text{(matrix product)} \end{aligned}$$

The second step writes out the **matrix product**. g' is continuous (Definition 0.7): $t \mapsto \tilde{z} + t\tilde{h}$ is continuous, since $\|(\tilde{z} + s\tilde{h}) - (\tilde{z} + t\tilde{h})\| = |s - t|\|\tilde{h}\|$, each $\frac{\partial}{\partial x_j} f$ is continuous by assumption, and compositions, sums and constant multiples of continuous functions are continuous. Each $\frac{\partial}{\partial x_j} f$ has continuous partial derivatives, the second partial derivatives of f , so the same chain-rule computation applies to it: $\frac{\partial}{\partial t} \left[\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right] = \sum_{i=1}^p h_i \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right)$. So

$$\begin{aligned} g''(t) &= \sum_{j=1}^p h_j \sum_{i=1}^p h_i \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right) && \text{(differentiate each term of } g'(t)) \\ &= \sum_{i=1}^p \sum_{j=1}^p h_i \left[\mathbf{H}_f(\tilde{z} + t\tilde{h}) \right]_{ij} h_j && \text{(Definition 6; reorder the finite sums)} \\ &= \tilde{h}^\top \mathbf{H}_f(\tilde{z} + t\tilde{h}) \tilde{h}. && \text{(matrix product)} \end{aligned}$$

The last step is again the **matrix product**. Now apply Theorem 0.14 to g on $[0, 1]$: there

is a $\tau \in (0, 1)$ with

$$\begin{aligned}
f(\tilde{z} + \tilde{h}) &= g(1) && \text{(definition of } g) \\
&= g(0) + g'(0)(1 - 0) + \frac{1}{2}g''(\tau)(1 - 0)^2 && \text{(Theorem 14, with } a = 0, b = 1) \\
&= g(0) + g'(0) + \frac{1}{2}g''(\tau) && (1 - 0 = 1) \\
&= f(\tilde{z}) + \tilde{h}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{z}) + \frac{1}{2}\tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau\tilde{h})\tilde{h} && \text{(substitute } g(0), g'(0), g''(\tau)) \\
&= f(\tilde{z}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) \right)^\top \tilde{h} + \frac{1}{2}\tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau\tilde{h})\tilde{h}. && \text{(a } 1 \times 1 \text{ matrix equals its transpose)}
\end{aligned}$$

□

Example 0.28 (For a quadratic form the expansion is exact). Let $f(\tilde{x}) = \tilde{x}^\top \mathbf{S}\tilde{x}$ with $\mathbf{S}$ symmetric and constant. Its gradient is $2\mathbf{S}\tilde{x}$ (Theorem 0.7) and its Hessian is $2\mathbf{S}$ at every point (Theorem 0.11), so whatever τ is, Theorem 0.15 reads

$$f(\tilde{z} + \tilde{h}) = \tilde{z}^\top \mathbf{S}\tilde{z} + (2\mathbf{S}\tilde{z})^\top \tilde{h} + \frac{1}{2}\tilde{h}^\top (2\mathbf{S})\tilde{h} = \tilde{z}^\top \mathbf{S}\tilde{z} + 2\tilde{z}^\top \mathbf{S}\tilde{h} + \tilde{h}^\top \mathbf{S}\tilde{h},$$

using $(2\mathbf{S}\tilde{z})^\top = 2\tilde{z}^\top \mathbf{S}^\top = 2\tilde{z}^\top \mathbf{S}$. Multiplying out directly gives the same:

$$\begin{aligned}
(\tilde{z} + \tilde{h})^\top \mathbf{S}(\tilde{z} + \tilde{h}) &= \tilde{z}^\top \mathbf{S}\tilde{z} + \tilde{z}^\top \mathbf{S}\tilde{h} + \tilde{h}^\top \mathbf{S}\tilde{z} + \tilde{h}^\top \mathbf{S}\tilde{h} && \text{(distribute)} \\
&= \tilde{z}^\top \mathbf{S}\tilde{z} + 2\tilde{z}^\top \mathbf{S}\tilde{h} + \tilde{h}^\top \mathbf{S}\tilde{h}. && (\tilde{h}^\top \mathbf{S}\tilde{z} = (\tilde{h}^\top \mathbf{S}\tilde{z})^\top = \tilde{z}^\top \mathbf{S}\tilde{h})
\end{aligned}$$

Definition 0.9 (Strict local minimizer). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. A point $\tilde{x}^*$ is a **strict local minimizer** of f if there is a number $\delta > 0$ such that $f(\tilde{x}^*) < f(\tilde{x})$ for every $\tilde{x} \neq \tilde{x}^*$ with $\|\tilde{x} - \tilde{x}^*\| < \delta$.

Example 0.29 (Strict and non-strict local minimizers).

- $f(x) = (x - 2)^2$ has $f(2) = 0 < (x - 2)^2 = f(x)$ for every $x \neq 2$, so 2 is a strict local minimizer (any $\delta > 0$ works).
- A constant function $f(\tilde{x}) = 0$ has $f(\tilde{x}^*) \leq f(\tilde{x})$ for all $\tilde{x}^*$ and $\tilde{x}$, so every point is a **local minimizer**, but no point is a strict one: $f(\tilde{x}^*) < f(\tilde{x})$ never holds.

Lemma 0.1 (A quadratic form is at least the smallest eigenvalue times the squared length). Let $\mathbf{A}$ be a symmetric $p \times p$ matrix, and let $\lambda_{\min}$ be the smallest of its eigenvalues

(*eigendecomposition*). Then $\tilde{h}^\top \mathbf{A} \tilde{h} \geq \lambda_{\min} \|\tilde{h}\|^2$ for every $\tilde{h} \in \mathbb{R}^p$.

i Proof

Proof. Write $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ with $\mathbf{Q}$ orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ (*spectral theorem*), and let $\tilde{y} \stackrel{\text{def}}{=} \mathbf{Q}^\top \tilde{h}$. Then

$$\begin{aligned} \tilde{h}^\top \mathbf{A} \tilde{h} &= \tilde{h}^\top \mathbf{Q}\mathbf{Q}^\top \tilde{h} && \text{(substitute the eigendecomposition)} \\ &= \tilde{y}^\top \tilde{y} && (\tilde{h}^\top \mathbf{Q} = (\mathbf{Q}^\top \tilde{h})^\top, \text{ transpose of a product}) \\ &= \sum_{i=1}^p \lambda_i y_i^2 && (\Lambda \text{ is diagonal}) \\ &\geq \sum_{i=1}^p \lambda_{\min} y_i^2 && (\lambda_i \geq \lambda_{\min} \text{ and } y_i^2 \geq 0) \\ &= \lambda_{\min} \|\tilde{y}\|^2 && \text{(the squared length is the sum of squares)} \\ &= \lambda_{\min} \|\tilde{h}\|^2. && \text{(orthogonal matrices preserve length)} \end{aligned}$$

The second step is the *transpose of a product*. The last step applies *orthogonal matrices preserve length* to $\mathbf{Q}^\top$, which is orthogonal because $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$ (*orthogonal matrix*). $\square$

Example 0.30 (Checking the bound). $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 (*eigenvalue example*), so $\lambda_{\min} = 1$. At $\tilde{h} = (1, 0)^\top$, $\tilde{h}^\top \mathbf{A} \tilde{h} = 2 \geq 1 \cdot 1$. At $\tilde{h} = (1, -1)^\top$, $\tilde{h}^\top \mathbf{A} \tilde{h} = 2 - 1 - 1 + 2 = 2 = 1 \cdot 2$: the bound holds with equality, since this $\tilde{h}$ is an eigenvector for $\lambda_{\min}$.

Lemma 0.2 (A quadratic form with small entries is small). Let $\mathbf{E}$ be a $p \times p$ matrix whose entries all satisfy $|e_{ij}| \leq m$. Then $|\tilde{h}^\top \mathbf{E} \tilde{h}| \leq pm \|\tilde{h}\|^2$ for every $\tilde{h} \in \mathbb{R}^p$.

i Proof

Proof. Let $\tilde{a} \stackrel{\text{def}}{=} (|h_1|, \dots, |h_p|)^\top$ and $\tilde{1} \stackrel{\text{def}}{=} (1, \dots, 1)^\top$, both in $\mathbb{R}^p$. Then

$$\begin{aligned}
|\tilde{h}^\top \mathbf{E} \tilde{h}| &= \left| \sum_{i=1}^p \sum_{j=1}^p h_i e_{ij} h_j \right| && \text{(matrix product)} \\
&\leq \sum_{i=1}^p \sum_{j=1}^p |h_i| |e_{ij}| |h_j| && \text{(triangle inequality for numbers)} \\
&\leq m \sum_{i=1}^p \sum_{j=1}^p |h_i| |h_j| && (|e_{ij}| \leq m) \\
&= m \left(\sum_{i=1}^p |h_i| \right)^2 && \text{(the double sum factors)} \\
&= m (\tilde{\mathbf{1}} \cdot \tilde{a})^2 && \text{(dot product)} \\
&\leq m \|\tilde{\mathbf{1}}\|^2 \|\tilde{a}\|^2 && \text{(Cauchy-Schwarz, squared)} \\
&= m p \|\tilde{h}\|^2. && (\|\tilde{\mathbf{1}}\|^2 = p \text{ and } \|\tilde{a}\|^2 = \sum_i h_i^2 = \|\tilde{h}\|^2)
\end{aligned}$$

The steps use the [matrix product](#), the [dot product](#) and the [Cauchy-Schwarz inequality](#). $\square$

Example 0.31 (Checking the bound). Let $\mathbf{E} = \begin{bmatrix} 0.1 & -0.2 \\ -0.2 & 0.1 \end{bmatrix}$, so $m = 0.2$ works, and $\tilde{h} = (1, 1)^\top$, so $\|\tilde{h}\|^2 = 2$. Then $\tilde{h}^\top \mathbf{E} \tilde{h} = 0.1 - 0.2 - 0.2 + 0.1 = -0.2$, and $|-0.2| = 0.2 \leq 2 \cdot 0.2 \cdot 2 = 0.8$.

Theorem 0.16 (Second-order sufficient condition). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ (Definition 0.7). If $\tilde{x}^*$ is a stationary point of f (Definition 0.8) and $\mathbf{H}_f(\tilde{x}^*)$ is [positive definite](#), then $\tilde{x}^*$ is a strict local minimizer of f (Definition 0.9).*

i Proof

Proof. $\mathbf{H}_f(\tilde{x}^*)$ is positive definite, and so symmetric, so its smallest eigenvalue λ is positive ([definiteness and eigenvalues](#)). Each of the p^2 entries of $\mathbf{H}_f$ is continuous at $\tilde{x}^*$ (Definition 0.7): for $\varepsilon = \lambda/(2p)$ there is a $\delta_{ij} > 0$ with $|\mathbf{H}_f(\tilde{y})_{ij} - \mathbf{H}_f(\tilde{x}^*)_{ij}| < \lambda/(2p)$ whenever $\|\tilde{y} - \tilde{x}^*\| < \delta_{ij}$. Let δ be the smallest of these p^2 numbers.

Now let $0 < \|\tilde{h}\| < \delta$. By Theorem 0.15 there is a $\tau \in (0, 1)$ for which the first step of the next display holds; let $\tilde{y} \stackrel{\text{def}}{=} \tilde{x}^* + \tau \tilde{h}$ and $\mathbf{E} \stackrel{\text{def}}{=} \mathbf{H}_f(\tilde{y}) - \mathbf{H}_f(\tilde{x}^*)$. Since $\|\tilde{y} - \tilde{x}^*\| = \tau \|\tilde{h}\| < \delta$, every entry of $\mathbf{E}$ has absolute value less than $\lambda/(2p)$, so Lemma 0.2 applies with $m = \lambda/(2p)$:

$$\begin{aligned}
f(\tilde{x}^* + \tilde{h}) - f(\tilde{x}^*) &= \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{x}^*) \right)^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{y}) \tilde{h} && \text{(Theorem 15)} \\
&= \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{y}) \tilde{h} && (\tilde{x}^* \text{ is stationary}) \\
&= \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{x}^*) \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{E} \tilde{h} && (\mathbf{H}_f(\tilde{y}) = \mathbf{H}_f(\tilde{x}^*) + \mathbf{E}; \text{ distribute}) \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 + \frac{1}{2} \tilde{h}^\top \mathbf{E} \tilde{h} && \text{(Lemma 1)} \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 - \frac{1}{2} |\tilde{h}^\top \mathbf{E} \tilde{h}| && \text{(a number is at least minus its absolute value)} \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 - \frac{1}{2} p \frac{\lambda}{2p} \|\tilde{h}\|^2 && \text{(Lemma 2, } m = \lambda/(2p)) \\
&= \frac{\lambda}{4} \|\tilde{h}\|^2 && \text{(arithmetic)} \\
&> 0. && (\lambda > 0 \text{ and } \tilde{h} \neq \tilde{0})
\end{aligned}$$

So $f(\tilde{x}^*) < f(\tilde{x})$ whenever $0 < \|\tilde{x} - \tilde{x}^*\| < \delta$ (take $\tilde{h} = \tilde{x} - \tilde{x}^*$). □

Example 0.32 (Classifying stationary points).

- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ has the stationary point $(1, 0)^\top$ (Example 0.25), and its Hessian is $\mathbf{H}_f(\tilde{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2\mathbf{I}_2$, which is positive definite, so $(1, 0)^\top$ is a strict local minimizer.
- $f(\tilde{x}) = x_1^2 - x_2^2$ has the stationary point $\tilde{0}$ and Hessian $\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$, which is not positive definite ($(0, 1)^\top$ gives -2), and indeed $\tilde{0}$ is not a local minimizer (Example 0.26).

Example 0.33 (The condition is sufficient, not necessary). $f(\tilde{x}) = x_1^2 + x_2^4$ and $g(\tilde{x}) = x_1^2 - x_2^4$ both have gradient $\tilde{0}$ at $\tilde{0}$ and the same Hessian there, $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, since $\frac{\partial}{\partial x_2} \left(\frac{\partial}{\partial x_2} (\pm x_2^4) \right) = \pm 12x_2^2 = 0$ at $x_2 = 0$. That Hessian is not positive definite ($(0, 1)^\top$ gives 0), so Theorem 0.16 says nothing about either function. In fact $\tilde{0}$ is a strict local minimizer of f , since $f(\tilde{x}) > 0 = f(\tilde{0})$ for $\tilde{x} \neq \tilde{0}$, but not a local minimizer of g , since $g(0, t) = -t^4 < 0 = g(\tilde{0})$ for $t \neq 0$.

Convexity in several variables

Source

This section is adapted from the convexity part of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source states the first-order test (in one direction) and the second-derivative test for convexity without proof; here both directions of the first-order test are proved, and the second-derivative test is proved from the Taylor theorem of Section . The source defines both concave and strictly concave as “ $-f$ is strictly convex”; here f is concave when $-f$ is convex. These parts of the source are not part of this section:

- its catalogs of convex sets and convex functions, including the positive semidefinite matrices and $-\log \det$
- intersections of infinitely many convex sets, and suprema of infinitely many convex functions
- its examples, exercises and figures, including the multivariate normal maximum likelihood example

Definition 0.10 (Convex set). A set $K \subseteq \mathbb{R}^p$ is **convex** if for all $\tilde{x}, \tilde{y} \in K$ and all $t \in [0, 1]$, the point $t\tilde{x} + (1-t)\tilde{y}$ is in K : K contains the whole line segment between any two of its points.

Example 0.34 (Convex sets).

- $\mathbb{R}^p$ is convex, and so is any set with at most one point.
- The closed ball $B = \{\tilde{x} : \|\tilde{x}\| \leq r\}$ is convex: for $\tilde{x}, \tilde{y} \in B$ and $t \in [0, 1]$,

$$\begin{aligned} \|t\tilde{x} + (1-t)\tilde{y}\| &\leq \|t\tilde{x}\| + \|(1-t)\tilde{y}\| && \text{(triangle inequality)} \\ &= t\|\tilde{x}\| + (1-t)\|\tilde{y}\| && \text{(norm of a multiple; } t, 1-t \geq 0) \\ &\leq tr + (1-t)r && (\tilde{x}, \tilde{y} \in B) \\ &= r. && \text{(arithmetic)} \end{aligned}$$

The first step is the [triangle inequality](#), and the second uses [the norm of a multiple](#).

Example 0.35 (A set that is not convex). $K = \{x \in \mathbb{R} : |x| \geq 1\}$ is not convex: -1 and 1 are in K , but with $t = \frac{1}{2}$ the point $\frac{1}{2}(-1) + \frac{1}{2}(1) = 0$ is not.

Theorem 0.17 (An intersection of convex sets is convex). *If K_1 and K_2 are convex subsets of $\mathbb{R}^p$ (Definition 0.10), then $K_1 \cap K_2$ is convex.*

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in K_1 \cap K_2$ and $t \in [0, 1]$. Then $\tilde{x}, \tilde{y} \in K_1$, so $t\tilde{x} + (1-t)\tilde{y} \in K_1$, since K_1 is convex; in the same way $t\tilde{x} + (1-t)\tilde{y} \in K_2$. So $t\tilde{x} + (1-t)\tilde{y} \in K_1 \cap K_2$. $\square$

Example 0.36 (Intersecting a ball with a half-plane). The closed unit ball in $\mathbb{R}^2$ is convex (Example 0.34), and so is the half-plane $\{\tilde{x} : x_1 \geq 0\}$: if $x_1 \geq 0$ and $y_1 \geq 0$, then $tx_1 + (1-t)y_1 \geq 0$ for $t \in [0, 1]$. By Theorem 0.17, the closed half-disk $\{\tilde{x} : \|\tilde{x}\| \leq 1, x_1 \geq 0\}$ is convex. A union need not be convex: $\{x \in \mathbb{R} : x \leq -1\}$ and $\{x \in \mathbb{R} : x \geq 1\}$ are convex, but their union is the set of Example 0.35.

Definition 0.11 (Strictly convex and concave functions). A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is **strictly convex** if $f(t\tilde{x} + (1-t)\tilde{y}) < tf(\tilde{x}) + (1-t)f(\tilde{y})$ for all $\tilde{x} \neq \tilde{y}$ in $\mathbb{R}^p$ and all $t \in (0, 1)$. It is **concave** if $-f$ is **convex**, and **strictly concave** if $-f$ is strictly convex.

Example 0.37 (Strictly convex, and convex but not strictly).

- $f(x) = x^2$ is strictly convex:

$$\begin{aligned} tx^2 + (1-t)y^2 - (tx + (1-t)y)^2 &= tx^2 + (1-t)y^2 - t^2x^2 - 2t(1-t)xy - (1-t)^2y^2 \\ &= t(1-t)x^2 - 2t(1-t)xy + t(1-t)y^2 \\ &= t(1-t)(x-y)^2, \end{aligned}$$

(expand the square)
($t - t^2 = t(1-t)$)
(factor)

which is positive when $x \neq y$ and $t \in (0, 1)$. At $t \in \{0, 1\}$ or $x = y$ it is 0, so x^2 is also convex.

- $f(x) = 2x + 1$ is convex but not strictly convex: $f(tx + (1-t)y) = 2tx + 2(1-t)y + 1 = tf(x) + (1-t)f(y)$ for every x, y, t , so the inequality holds, but never strictly. It is also concave.

Theorem 0.18 (The maximum of convex functions is convex). *If f_1 and f_2 are convex functions on $\mathbb{R}^p$, then $f(\tilde{x}) \stackrel{\text{def}}{=} \max\{f_1(\tilde{x}), f_2(\tilde{x})\}$ is convex.*

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ and $t \in [0, 1]$. For $i = 1, 2$,

$$\begin{aligned} f_i(t\tilde{x} + (1-t)\tilde{y}) &\leq tf_i(\tilde{x}) + (1-t)f_i(\tilde{y}) \quad (f_i \text{ is convex}) \\ &\leq tf(\tilde{x}) + (1-t)f(\tilde{y}). \quad (f_i \leq f, \text{ and } t, 1-t \geq 0) \end{aligned}$$

Both f_1 and f_2 at $t\tilde{x} + (1-t)\tilde{y}$ are at most $tf(\tilde{x}) + (1-t)f(\tilde{y})$, so their maximum $f(t\tilde{x} + (1-t)\tilde{y})$ is too. $\square$

Example 0.38 (A maximum is convex; a minimum need not be).

- $f_1(x) = x$ is convex, by the computation for $2x + 1$ in Example 0.37 with 2 and 1 replaced by 1 and 0, and $f_2(x) = x^2$ is convex by the same example, so $\max\{x, x^2\}$ is convex by Theorem 0.18.
- The minimum of two convex functions need not be convex: $g(x) = \min\{x^2, (x-2)^2\}$ has $g(0) = g(2) = 0$, but at the midpoint $g(1) = \min\{1, 1\} = 1 > \frac{1}{2}g(0) + \frac{1}{2}g(2) = 0$.

Definition 0.12 (Epigraph). The **epigraph** of $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is the set of points on or over its graph,

$$\text{epi } f \stackrel{\text{def}}{=} \{(\tilde{x}, s) : \tilde{x} \in \mathbb{R}^p, s \in \mathbb{R}, f(\tilde{x}) \leq s\}.$$

A pair $(\tilde{x}, s)$ is treated as the $(p+1)$ -vector with entries $x_1, \dots, x_p, s$, so $\text{epi } f \subseteq \mathbb{R}^{p+1}$, and sums and multiples of pairs are taken entry by entry.

Example 0.39 (The epigraph of x^2). For $f(x) = x^2$, $\text{epi } f = \{(x, s) : x^2 \leq s\}$, the points (x, s) on the parabola $s = x^2$ or on the side of it where s is larger. $(1, 2)$ is in it, since $1^2 \leq 2$; $(2, 1)$ is not, since $2^2 > 1$.

Theorem 0.19 (Convexity through the epigraph). $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is **convex** if and only if its epigraph (Definition 0.12) is convex (Definition 0.10).

i Proof

Proof. If. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ and $t \in [0, 1]$. The points $(\tilde{x}, f(\tilde{x}))$ and $(\tilde{y}, f(\tilde{y}))$ are in $\text{epi } f$, since $f(\tilde{x}) \leq f(\tilde{x})$ and $f(\tilde{y}) \leq f(\tilde{y})$. Since $\text{epi } f$ is convex, it contains

$$t(\tilde{x}, f(\tilde{x})) + (1-t)(\tilde{y}, f(\tilde{y})) = (t\tilde{x} + (1-t)\tilde{y}, tf(\tilde{x}) + (1-t)f(\tilde{y})),$$

and by Definition 0.12 that membership means $f(t\tilde{x} + (1-t)\tilde{y}) \leq tf(\tilde{x}) + (1-t)f(\tilde{y})$.

Only if. Let $(\tilde{x}, s)$ and $(\tilde{y}, u)$ be in $\text{epi } f$, so $f(\tilde{x}) \leq s$ and $f(\tilde{y}) \leq u$, and let $t \in [0, 1]$. Then

$$\begin{aligned} f(t\tilde{x} + (1-t)\tilde{y}) &\leq tf(\tilde{x}) + (1-t)f(\tilde{y}) \quad (f \text{ is convex}) \\ &\leq ts + (1-t)u, \quad (f(\tilde{x}) \leq s, f(\tilde{y}) \leq u, \text{ and } t, 1-t \geq 0) \end{aligned}$$

so $t(\tilde{x}, s) + (1-t)(\tilde{y}, u) = (t\tilde{x} + (1-t)\tilde{y}, ts + (1-t)u)$ is in $\text{epi } f$. $\square$

Example 0.40 (Two epigraphs).

- The epigraph of $f(x) = |x|$ is $\{(x, s) : |x| \leq s\}$, a wedge. It is the intersection of the half-planes $\{(x, s) : s - x \geq 0\}$ and $\{(x, s) : s + x \geq 0\}$. Each half-plane is convex: if $s_1 - x_1 \geq 0$ and $s_2 - x_2 \geq 0$, then $(ts_1 + (1-t)s_2) - (tx_1 + (1-t)x_2) = t(s_1 - x_1) + (1-t)(s_2 - x_2) \geq 0$, and the same with $+$ for the other. So the wedge is convex (Theorem 0.17), and $|x|$ is convex by Theorem 0.19.
- The epigraph of $g(x) = -x^2$ is not convex: $(-1, -1)$ and $(1, -1)$ are in it, since $g(\pm 1) = -1$, but their midpoint $(0, -1)$ is not, since $g(0) = 0 > -1$. So $-x^2$ is not convex.

Theorem 0.20 (First-order characterization of convexity). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (Definition 0.7). Then f is **convex** if and only if*

$$f(\tilde{x}) \geq f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}) \quad \text{for all } \tilde{x}, \tilde{y} \in \mathbb{R}^p : \quad (1)$$

every tangent plane lies on or under the graph. If the inequality is strict whenever $\tilde{x} \neq \tilde{y}$, then f is strictly convex (Definition 0.11).

i Proof

Proof. **Only if.** Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$, let $\tilde{d} \stackrel{\text{def}}{=} \tilde{x} - \tilde{y}$, and let $g(t) \stackrel{\text{def}}{=} f(\tilde{y} + t\tilde{d})$. The inner function $t \mapsto \tilde{y} + t\tilde{d}$ has derivative $\tilde{d}^\top$ (Definition 0.3), so the vector chain rule (Theorem 0.8) gives

$$\begin{aligned} g'(0) &= \tilde{d}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \quad (\text{Theorem 8}) \\ &= \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top \tilde{d}. \quad (\text{a } 1 \times 1 \text{ matrix equals its transpose}) \end{aligned}$$

For $t \in (0, 1]$, $\tilde{y} + t\tilde{d} = t\tilde{x} + (1-t)\tilde{y}$, so

$$\begin{aligned}
g(t) - g(0) &= f(t\tilde{x} + (1-t)\tilde{y}) - f(\tilde{y}) && \text{(definition of } g) \\
&\leq tf(\tilde{x}) + (1-t)f(\tilde{y}) - f(\tilde{y}) && (f \text{ is convex}) \\
&= t(f(\tilde{x}) - f(\tilde{y})), && \text{(collect terms)}
\end{aligned}$$

and dividing by $t > 0$, $\frac{g(t) - g(0)}{t} \leq f(\tilde{x}) - f(\tilde{y})$. As $t \rightarrow 0$ from the right, the left side tends to $g'(0)$ (g is differentiable at 0, so the one-sided limit equals the two-sided derivative), and a limit of numbers that are all at most $f(\tilde{x}) - f(\tilde{y})$ is at most $f(\tilde{x}) - f(\tilde{y})$. So $(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}))^\top (\tilde{x} - \tilde{y}) \leq f(\tilde{x}) - f(\tilde{y})$, which rearranges to Equation 1.

If. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$, $t \in [0, 1]$, $\tilde{z} \stackrel{\text{def}}{=} t\tilde{x} + (1-t)\tilde{y}$, and $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{z})$. Applying Equation 1 at the point $\tilde{z}$, once toward $\tilde{x}$ and once toward $\tilde{y}$,

$$f(\tilde{x}) \geq f(\tilde{z}) + \tilde{g}^\top (\tilde{x} - \tilde{z}), \quad f(\tilde{y}) \geq f(\tilde{z}) + \tilde{g}^\top (\tilde{y} - \tilde{z}).$$

Multiplying the first by $t \geq 0$ and the second by $1 - t \geq 0$ and adding,

$$\begin{aligned}
tf(\tilde{x}) + (1-t)f(\tilde{y}) &\geq t(f(\tilde{z}) + \tilde{g}^\top (\tilde{x} - \tilde{z})) + (1-t)(f(\tilde{z}) + \tilde{g}^\top (\tilde{y} - \tilde{z})) && \text{(multiply by } t, 1-t \geq 0 \text{ and add)} \\
&= (t + (1-t))f(\tilde{z}) + t\tilde{g}^\top (\tilde{x} - \tilde{z}) + (1-t)\tilde{g}^\top (\tilde{y} - \tilde{z}) && \text{(distribute } t \text{ and } 1-t) \\
&= f(\tilde{z}) + t\tilde{g}^\top (\tilde{x} - \tilde{z}) + (1-t)\tilde{g}^\top (\tilde{y} - \tilde{z}) && (t + (1-t) = 1) \\
&= f(\tilde{z}) + \tilde{g}^\top (t(\tilde{x} - \tilde{z}) + (1-t)(\tilde{y} - \tilde{z})) && \text{(factor out } \tilde{g}^\top; \text{ matrix products distributive)} \\
&= f(\tilde{z}) + \tilde{g}^\top (t\tilde{x} + (1-t)\tilde{y} - \tilde{z}) && \text{(collect terms)} \\
&= f(\tilde{z}). && \text{(definition of } \tilde{z})
\end{aligned}$$

That inequality is the defining inequality of a **convex function**. If Equation 1 is strict for distinct points, and $\tilde{x} \neq \tilde{y}$ and $t \in (0, 1)$, then:

- $\tilde{z}$ differs from both $\tilde{x}$ and $\tilde{y}$, since $\tilde{x} - \tilde{z} = (1-t)(\tilde{x} - \tilde{y}) \neq \tilde{0}$ and $\tilde{y} - \tilde{z} = t(\tilde{y} - \tilde{x}) \neq \tilde{0}$;
- so both inequalities at $\tilde{z}$ are strict;
- both multipliers t and $1 - t$ are positive, so the first step of the display is strict, and f is strictly convex.

□

Example 0.41 (Tangent lines under a parabola). For $f(x) = x^2$, the right side of Equation 1 is $y^2 + 2y(x - y) = 2xy - y^2$, and

$$f(x) - (2xy - y^2) = x^2 - 2xy + y^2 = (x - y)^2 \geq 0,$$

with equality only at $x = y$, confirming that x^2 is strictly convex. For $g(x) = -x^2$ the inequality fails: at $y = 0$ the tangent line is $s = 0$, and $g(1) = -1 < 0$.

Theorem 0.21 (Second-derivative test for convexity). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ (Definition 0.7).*

1. *If $\mathbf{H}_f(\tilde{x})$ is **positive semidefinite** for every $\tilde{x}$, then f is **convex**.*
2. *If $\mathbf{H}_f(\tilde{x})$ is **positive definite** for every $\tilde{x}$, then f is **strictly convex** (Definition 0.11).*

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$. By Theorem 0.15 with $\tilde{z} = \tilde{y}$ and $\tilde{h} = \tilde{x} - \tilde{y}$, there is a $\tau \in (0, 1)$ such that, with $\tilde{w} \stackrel{\text{def}}{=} \tilde{y} + \tau(\tilde{x} - \tilde{y})$,

$$\begin{aligned} f(\tilde{x}) &= f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}) + \frac{1}{2} (\tilde{x} - \tilde{y})^\top \mathbf{H}_f(\tilde{w}) (\tilde{x} - \tilde{y}) \quad (\text{Theorem 15}) \\ &\geq f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}). \end{aligned} \quad (\mathbf{H}_f(\tilde{w}) \text{ is positive semidefinite})$$

This inequality is Equation 1, so f is convex by Theorem 0.20. Under the hypothesis of part 2 and with $\tilde{x} \neq \tilde{y}$, the quadratic-form term is positive, so the inequality is strict, and f is strictly convex by the last sentence of Theorem 0.20. $\square$

Example 0.42 (Using the Hessian to show convexity).

- For a symmetric positive semidefinite $\mathbf{A}$, $f(\tilde{x}) = \tilde{x}^\top \mathbf{A} \tilde{x}$ has Hessian $2\mathbf{A}$ (Theorem 0.11), which is positive semidefinite because $\tilde{h}^\top (2\mathbf{A}) \tilde{h} = 2\tilde{h}^\top \mathbf{A} \tilde{h} \geq 0$; so f is convex.
- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ has Hessian $2\mathbf{I}_2$ (Example 0.32), which is positive definite, so f is strictly convex.
- $f(\tilde{x}) = e^{2x_1+x_2} - x_1$ has Hessian $e^u \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ with $u = 2x_1 + x_2$ (Example 0.18), and

$$\tilde{h}^\top \mathbf{H}_f(\tilde{x}) \tilde{h} = e^u (4h_1^2 + 4h_1h_2 + h_2^2) = e^u (2h_1 + h_2)^2 \geq 0,$$

so f is convex. The Hessian is not positive definite ($\tilde{h} = (1, -2)^\top$ gives 0), so part 2 does not apply, and in fact f is not strictly convex: along $\tilde{x} = t(1, -2)^\top$, $u = 2t - 2t = 0$ and $f = 1 - t$ is linear in t , so the convexity inequality holds with equality there.

Example 0.43 (Strictly convex with a singular Hessian). Part 2 is sufficient but not necessary. $f(x) = x^4$ has $f''(0) = 12 \cdot 0^2 = 0$, so $\mathbf{H}_f(0) = [0]$ is not positive definite. Yet f is strictly convex, by the strict form of Equation 1: for $x \neq y$,

$$\begin{aligned} x^4 - y^4 - 4y^3(x - y) &= (x - y)(x^3 + x^2y + xy^2 + y^3) - 4y^3(x - y) && \text{(factor } x^4 - y^4) \\ &= (x - y)(x^3 + x^2y + xy^2 - 3y^3) && \text{(collect the } y^3 \text{ terms)} \\ &= (x - y)^2(x^2 + 2xy + 3y^2) && ((x - y)(x^2 + 2xy + 3y^2) = x^3 + x^2y + xy^2 - 3y^3) \\ &= (x - y)^2((x + y)^2 + 2y^2), && \text{(complete the square)} \end{aligned}$$

which is positive: $(x - y)^2 > 0$, and $(x + y)^2 + 2y^2 = 0$ would need $y = 0$ and then $x = 0 = y$.

Corollary 0.4 (Stationary points of a convex function are global minimizers). *If f is as in Theorem 0.20 and convex, then every stationary point $\tilde{x}^*$ of f (Definition 0.8) is a global minimizer.*

i Proof

Proof. For every $\tilde{x}$,

$$\begin{aligned} f(\tilde{x}) &\geq f(\tilde{x}^*) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{x}^*) \right)^\top (\tilde{x} - \tilde{x}^*) && \text{(Equation 1 with } \tilde{y} = \tilde{x}^*) \\ &= f(\tilde{x}^*) + \tilde{0}^\top (\tilde{x} - \tilde{x}^*) && (\tilde{x}^* \text{ is stationary}) \\ &= f(\tilde{x}^*). && (\tilde{0}^\top \tilde{v} = 0) \end{aligned}$$

□

Example 0.44 (A convex function, and a non-convex one).

- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ is convex by Theorem 0.21, since its Hessian is $2\mathbf{I}_2$ (Example 0.32), so its stationary point $(1, 0)^\top$ (Example 0.25) is a global minimizer, as completing the square showed in Example 0.26.
- Without convexity, a stationary point need not be a minimizer at all: $x_1^2 - x_2^2$ has the stationary point $\tilde{0}$, which is not a minimizer (Example 0.26).
- The corollary says nothing about whether a minimizer exists: $e^{2x_1+x_2} - x_1$ is convex (Example 0.42) but has no stationary point, and so no local minimizer (Example 0.26).

Minimizing a quadratic under linear constraints

Source

This section is adapted from the Lagrange-multiplier part of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source solves the stationarity equations of the Lagrangian and reports the solution; here the solution is also proved to be the unique minimizer, directly from the positive definiteness of the matrices involved. The source assumes only that $\mathbf{S}$ is positive definite; the added hypothesis that the columns of $\mathbf{A}$ are linearly independent makes $\mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ invertible. The source's example with general coefficients a_1, a_2, b is specialized here to 3, 4, 5. These parts of the source are not part of this section:

- the Lagrange multiplier method for constraints that are not linear, or objectives that are not quadratic, including its multinomial maximum likelihood example
- the eigenvalues of the block matrix of the stationarity equations, and the convexity of the Lagrangian in $\tilde{x}$ and its concavity in the multipliers
- its homework examples

Here the objective is a quadratic $\frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ with $\mathbf{S}$ positive definite, and the constraints are m linear equations $\mathbf{A}^\top \tilde{x} = \tilde{b}$, where $\mathbf{A}$ is a $p \times m$ matrix and $\tilde{b} \in \mathbb{R}^m$. The only stationary point of the objective alone (Definition 0.8) is $\tilde{0}_{p \times 1}$ (its gradient is $\mathbf{S} \tilde{x}$, and $\mathbf{S}$ is invertible), which may not satisfy the constraints. A Lagrange multiplier adds one unknown per constraint, and the constrained minimizer is then found among the stationary points of a new function of the extended set of unknowns.

Definition 0.13 (Lagrangian). For the problem of minimizing $f(\tilde{x})$ over $\tilde{x} \in \mathbb{R}^p$ subject to $\mathbf{A}^\top \tilde{x} = \tilde{b}$, with $\mathbf{A}$ a $p \times m$ matrix and $\tilde{b} \in \mathbb{R}^m$, the **Lagrangian** is the function of $\tilde{x} \in \mathbb{R}^p$ and $\tilde{\lambda} \in \mathbb{R}^m$

$$L(\tilde{x}, \tilde{\lambda}) \stackrel{\text{def}}{=} f(\tilde{x}) + \tilde{\lambda}^\top (\mathbf{A}^\top \tilde{x} - \tilde{b}),$$

and the entries of $\tilde{\lambda}$ are the **Lagrange multipliers**, one per constraint.

Example 0.45 (The gradients of the Lagrangian of a quadratic). Let $f(\tilde{x}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ with $\mathbf{S}$ symmetric and constant. By the [transpose of a product](#), $\tilde{\lambda}^\top \mathbf{A}^\top = (\mathbf{A} \tilde{\lambda})^\top$, so

$$\begin{aligned} L(\tilde{x}, \tilde{\lambda}) &= \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + \tilde{\lambda}^\top \mathbf{A}^\top \tilde{x} - \tilde{\lambda}^\top \tilde{b} \quad (\text{distribute}) \\ &= \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + (\mathbf{A} \tilde{\lambda})^\top \tilde{x} - \tilde{\lambda}^\top \tilde{b}, \quad (\text{transpose of a product}) \end{aligned}$$

and, since $\tilde{\lambda}^\top (\mathbf{A}^\top \tilde{x} - \tilde{b})$ is 1×1 and so equals its transpose,

$$L(\tilde{x}, \tilde{\lambda}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + (\mathbf{A}^\top \tilde{x} - \tilde{b})^\top \tilde{\lambda}.$$

The partial derivatives of a sum are the sums of the partial derivatives, entry by entry. Using the first of these forms, Theorem 0.7 for the first term, Corollary 0.1 for the second ($\mathbf{A}\tilde{\lambda}$ is constant in $\tilde{x}$), and Definition 0.4 for the third,

$$\begin{aligned} \frac{\partial}{\partial \tilde{x}} L(\tilde{x}, \tilde{\lambda}) &= \frac{1}{2} \frac{\partial}{\partial \tilde{x}} (\tilde{x}^\top \mathbf{S} \tilde{x}) + \frac{\partial}{\partial \tilde{x}} ((\mathbf{A}\tilde{\lambda})^\top \tilde{x}) - \frac{\partial}{\partial \tilde{x}} (\tilde{\lambda}^\top \tilde{b}) \quad (\text{sum and constant multiple}) \\ &= \frac{1}{2} (2\mathbf{S}\tilde{x}) + \mathbf{A}\tilde{\lambda} - \tilde{0}_{p \times 1} \quad (\text{the three derivatives}) \\ &= \mathbf{S}\tilde{x} + \mathbf{A}\tilde{\lambda}. \quad (\text{simplify}) \end{aligned}$$

Using the second form, in which $f(\tilde{x})$ and $\mathbf{A}^\top \tilde{x} - \tilde{b}$ are constant in $\tilde{\lambda}$, and Corollary 0.1 again,

$$\begin{aligned} \frac{\partial}{\partial \tilde{\lambda}} L(\tilde{x}, \tilde{\lambda}) &= \frac{\partial}{\partial \tilde{\lambda}} f(\tilde{x}) + \frac{\partial}{\partial \tilde{\lambda}} ((\mathbf{A}^\top \tilde{x} - \tilde{b})^\top \tilde{\lambda}) \quad (\text{sum}) \\ &= \tilde{0}_{m \times 1} + (\mathbf{A}^\top \tilde{x} - \tilde{b}) \quad (\text{the two derivatives}) \\ &= \mathbf{A}^\top \tilde{x} - \tilde{b}. \quad (\text{simplify}) \end{aligned}$$

So $(\tilde{x}, \tilde{\lambda})$ makes the gradients equal $\tilde{0}_{p \times 1}$ and $\tilde{0}_{m \times 1}$ exactly when

$$\mathbf{S}\tilde{x} + \mathbf{A}\tilde{\lambda} = \tilde{0}_{p \times 1} \quad \text{and} \quad \mathbf{A}^\top \tilde{x} = \tilde{b}, \quad \text{that is,} \quad \underbrace{\begin{bmatrix} \mathbf{S} & \mathbf{A} \\ \mathbf{A}^\top & \tilde{0}_{m \times m} \end{bmatrix}}_{(p+m) \times (p+m)} \underbrace{\begin{bmatrix} \tilde{x} \\ \tilde{\lambda} \end{bmatrix}}_{(p+m) \times 1} = \underbrace{\begin{bmatrix} \tilde{0}_{p \times 1} \\ \tilde{b} \end{bmatrix}}_{(p+m) \times 1}, \quad (2)$$

the second equation being the constraint itself.

Theorem 0.22 (Minimizing a positive definite quadratic under linear constraints). *Let $\mathbf{S}$ be a $p \times p$ **positive definite** matrix, $\mathbf{A}$ a $p \times m$ matrix with **linearly independent** columns, and $\tilde{b} \in \mathbb{R}^m$. Then the $m \times m$ matrix $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ is invertible, and $f(\tilde{x}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ has exactly one minimizer over the $\tilde{x} \in \mathbb{R}^p$ with $\mathbf{A}^\top \tilde{x} = \tilde{b}$, namely*

$$\tilde{x}^* = \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b}.$$

With $\tilde{\lambda}^ \stackrel{\text{def}}{=} -\mathbf{M}^{-1} \tilde{b}$, the pair $(\tilde{x}^*, \tilde{\lambda}^*)$ solves Equation 2, and the minimum value is $f(\tilde{x}^*) = \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \tilde{b}$.*

i Proof

Proof. The steps marked “transpose of a product” use the [transpose of a product](#), and $\mathbf{S}$ is symmetric because it is [positive definite](#).

$\mathbf{M}$ is invertible. $\mathbf{S}^{-1}$ exists and is positive definite ([positive definite inverse](#)), so $\mathbf{M} = \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ is positive definite ([operations that preserve definiteness](#), part 1), and so invertible (positive definite inverse again). $\mathbf{S}^{-1}$ and $\mathbf{M}^{-1}$ are symmetric ([inverse of a symmetric matrix](#)), because $\mathbf{S}$ and $\mathbf{M}$ are.

$\tilde{x}^*$ satisfies the constraint, and the pair solves Equation 2.

$$\begin{aligned}
 \mathbf{A}^\top \tilde{x}^* &= \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^*) \\
 &= \mathbf{M} \mathbf{M}^{-1} \tilde{b} && \text{(definition of } \mathbf{M}) \\
 &= \tilde{b}, && (\mathbf{M} \mathbf{M}^{-1} = \mathbf{I}_m) \\
 \mathbf{S} \tilde{x}^* + \mathbf{A} \tilde{\lambda}^* &= \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} - \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^* \text{ and } \tilde{\lambda}^*) \\
 &= \mathbf{A} \mathbf{M}^{-1} \tilde{b} - \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S} \mathbf{S}^{-1} = \mathbf{I}_p) \\
 &= \tilde{0}_{p \times 1}. && \text{(subtract)}
 \end{aligned}$$

$\tilde{x}^*$ is the unique minimizer. Let $\tilde{x}$ satisfy $\mathbf{A}^\top \tilde{x} = \tilde{b}$, and let $\tilde{d} \stackrel{\text{def}}{=} \tilde{x} - \tilde{x}^*$, so $\mathbf{A}^\top \tilde{d} = \mathbf{A}^\top \tilde{x} - \mathbf{A}^\top \tilde{x}^* = \tilde{b} - \tilde{b} = \tilde{0}_{m \times 1}$ (matrix products distribute; both points satisfy the constraint). First, the cross term vanishes:

$$\begin{aligned}
 \tilde{d}^\top \mathbf{S} \tilde{x}^* &= \tilde{d}^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^*) \\
 &= \tilde{d}^\top \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S} \mathbf{S}^{-1} = \mathbf{I}_p) \\
 &= (\mathbf{A}^\top \tilde{d})^\top \mathbf{M}^{-1} \tilde{b} && \text{(transpose of a product; } (\mathbf{A}^\top)^\top = \mathbf{A}) \\
 &= \tilde{0}_{m \times 1}^\top \mathbf{M}^{-1} \tilde{b} && (\mathbf{A}^\top \tilde{d} = \tilde{0}_{m \times 1}) \\
 &= 0. && \text{(a product with a zero factor is zero)}
 \end{aligned}$$

Then

$$\begin{aligned}
f(\tilde{x}) &= \frac{1}{2} (\tilde{x}^* + \tilde{d})^\top \mathbf{S} (\tilde{x}^* + \tilde{d}) & (\tilde{x} = \tilde{x}^* + \tilde{d}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} + \tilde{d}^\top) \mathbf{S} (\tilde{x}^* + \tilde{d}) & (\text{the transpose of a sum is the sum of the transposes}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{x}^{*\top} \mathbf{S} \tilde{d} + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) & (\text{distribute}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S}^\top \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) & (1 \times 1: \tilde{x}^{*\top} \mathbf{S} \tilde{d} = (\tilde{x}^* \mathbf{S} \tilde{d})^\top; \text{transpose of a product}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + 2 \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) & (\mathbf{S}^\top = \mathbf{S}; \text{combine}) \\
&= \frac{1}{2} \tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \frac{1}{2} \tilde{d}^\top \mathbf{S} \tilde{d} & (\text{distribute } \frac{1}{2}) \\
&= f(\tilde{x}^*) + \frac{1}{2} \tilde{d}^\top \mathbf{S} \tilde{d} & (\text{the cross term is 0}) \\
&\geq f(\tilde{x}^*), & (\mathbf{S} \text{ is positive definite})
\end{aligned}$$

with equality only when $\tilde{d} = \tilde{0}_{p \times 1}$, that is, $\tilde{x} = \tilde{x}^*$.

The minimum value.

$$\begin{aligned}
f(\tilde{x}^*) &= \frac{1}{2} (\mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b})^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} & (\text{substitute } \tilde{x}^*) \\
&= \frac{1}{2} \tilde{b}^\top (\mathbf{M}^{-1})^\top \mathbf{A}^\top (\mathbf{S}^{-1})^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} & (\text{transpose of a product, applied repeatedly}) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} & (\mathbf{S}^{-1} \text{ and } \mathbf{M}^{-1} \text{ are symmetric}) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} & (\mathbf{S}^{-1} \mathbf{S} = \mathbf{I}_p) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{M} \mathbf{M}^{-1} \tilde{b} & (\text{definition of } \mathbf{M}) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \tilde{b}. & (\mathbf{M}^{-1} \mathbf{M} = \mathbf{I}_m)
\end{aligned}$$

□

Example 0.46 (The closest point on a line to the origin). Minimize $x_1^2 + x_2^2$ subject to $3x_1 + 4x_2 = 5$. In the form of Theorem 0.22, $x_1^2 + x_2^2 = \frac{1}{2} \tilde{x}^\top (2\mathbf{I}_2) \tilde{x}$, so $\mathbf{S} = 2\mathbf{I}_2$, $\mathbf{A} = (3, 4)^\top$ ($p = 2$, $m = 1$) and $\tilde{b} = [5]$. The hypotheses hold: $2\mathbf{I}_2$ is positive definite, and the single column $(3, 4)^\top$ is nonzero, so it is linearly independent. Also $\mathbf{S}^{-1} = \frac{1}{2}\mathbf{I}_2$. Then, from the formulas of Theorem 0.22,

$$\begin{aligned}
\mathbf{M} &= \mathbf{A}^\top \frac{1}{2} \mathbf{I}_2 \mathbf{A} = \frac{1}{2} (3^2 + 4^2) = \frac{25}{2}, \\
\tilde{x}^* &= \frac{1}{2} \mathbf{I}_2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} \left(\frac{2}{25} \right) (5) = \frac{5}{25} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}, \\
\tilde{\lambda}^* &= -\left(\frac{2}{25} \right) (5) = -0.4, \\
f(\tilde{x}^*) &= \frac{1}{2} (5) \left(\frac{2}{25} \right) (5) = 1.
\end{aligned}$$

Checks:

- the constraint: $3(0.6) + 4(0.8) = 1.8 + 3.2 = 5$;
- the first block of Equation 2: $\mathbf{S}\tilde{\mathbf{x}}^* + \mathbf{A}\tilde{\lambda}^* = (1.2, 1.6)^\top + (3, 4)^\top(-0.4) = (1.2, 1.6)^\top - (1.2, 1.6)^\top = \tilde{\mathbf{0}}_{2 \times 1}$;
- the value: $0.6^2 + 0.8^2 = 0.36 + 0.64 = 1$.

Geometrically, $\tilde{\mathbf{x}}^*$ is the point of the line nearest the origin, at distance 1.

Example 0.47 (The hypotheses are needed).

- If $\mathbf{S}$ is not positive definite there may be no minimizer: with $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and the constraint $x_1 = 1$ ($\mathbf{A} = (1, 0)^\top$, $\tilde{\mathbf{b}} = [1]$), $f(1, x_2) = \frac{1}{2}(1 - x_2^2)$ decreases without bound as x_2 grows.
- If the columns of $\mathbf{A}$ are dependent, $\mathbf{M}$ is not invertible: with $\mathbf{S} = \mathbf{I}_2$ and $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{M} = \mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, whose columns are equal. So $\mathbf{M}(1, -1)^\top = \tilde{\mathbf{0}}_{2 \times 1}$, and $\mathbf{M}$ is not invertible (a square matrix is invertible exactly when its null space is zero). The two constraints both read $x_1 + x_2 = b_i$, so they have no solution at all when $b_1 \neq b_2$. When $b_1 = b_2$ a unique minimizer still exists, $x_1 = x_2 = b_1/2$, but the formula of Theorem 0.22 cannot produce it.

Corollary 0.5 (The multiplier is the sensitivity of the minimum value). In Theorem 0.22, write the minimum value as a function of $\tilde{\mathbf{b}}$, $f^*(\tilde{\mathbf{b}}) = \frac{1}{2} \tilde{\mathbf{b}}^\top \mathbf{M}^{-1} \tilde{\mathbf{b}}$. Then

$$\frac{\partial}{\partial \tilde{\mathbf{b}}} f^*(\tilde{\mathbf{b}}) = \mathbf{M}^{-1} \tilde{\mathbf{b}} = -\tilde{\lambda}^*.$$

i Proof

Proof. $\mathbf{M}^{-1}$ is symmetric and does not depend on $\tilde{\mathbf{b}}$, so Theorem 0.7 applies with $\mathbf{M}^{-1}$ in place of $\mathbf{S}$:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\mathbf{b}}} f^*(\tilde{\mathbf{b}}) &= \frac{1}{2} \frac{\partial}{\partial \tilde{\mathbf{b}}} (\tilde{\mathbf{b}}^\top \mathbf{M}^{-1} \tilde{\mathbf{b}}) && \text{(constant multiple)} \\ &= \frac{1}{2} (2\mathbf{M}^{-1} \tilde{\mathbf{b}}) && \text{(derivative of a quadratic form)} \\ &= \mathbf{M}^{-1} \tilde{\mathbf{b}} && \text{(simplify)} \\ &= -\tilde{\lambda}^*. && \text{(definition of } \tilde{\lambda}^*) \end{aligned}$$

□

Example 0.48 (Moving the line). In Example 0.46 with a general right side b in place of 5, $\mathbf{M}^{-1} = \frac{2}{25}$ and $f^*(b) = \frac{1}{2}(\frac{2}{25})b^2 = \frac{b^2}{25}$, so $\frac{d}{db}f^*(b) = \frac{2b}{25}$. At $b = 5$ this derivative is $\frac{10}{25} = 0.4 = -\lambda^*$: moving the line $3x_1 + 4x_2 = b$ outward by a small step ε in b raises the minimum value by about 0.4ε .

Newton's method and gradient descent

Source

This section is adapted from the Newton-Raphson and gradient descent parts of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source motivates the Newton step by minimizing a quadratic approximation; here that step is proved to be the exact minimizer of the quadratic model. In the source's zigzag example the starting value of the objective is $\frac{1}{2}(b^2 + b)$, not $\frac{1}{2}(1 + b^2)$ as the source writes; the ratio of successive values is unaffected. In the same example the source's formula for $\tilde{x}^{(1)}$ has $\tilde{x}^{(1)}$ on its right side where $\tilde{x}^{(0)}$ is meant. These parts of the source are not part of this section:

- the quadratic convergence theorem for Newton's method in general, and its remedies for instability (a positive definite substitute for the Hessian, and line search)
- its remark that Hessians are expensive to compute
- the convergence-rate theorem for strongly convex functions
- gradient descent with momentum, and its optimal parameters
- Nesterov acceleration, and stochastic gradient descent
- its code and animations

Definition 0.14 (Newton's method for minimization). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives, and let $\tilde{x}^{(0)}$ be a starting point. **Newton's method** computes, for $t = 0, 1, 2, \dots$, as long as $\mathbf{H}_f(\tilde{x}^{(t)})$ is invertible,

$$\tilde{x}^{(t+1)} \stackrel{\text{def}}{=} \tilde{x}^{(t)} - [\mathbf{H}_f(\tilde{x}^{(t)})]^{-1} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}).$$

The vector $-[\mathbf{H}_f(\tilde{x}^{(t)})]^{-1} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)})$ is the **Newton step** at $\tilde{x}^{(t)}$.

Example 0.49 (Newton's method for a cubic). Let $f(x) = \frac{1}{3}x^3 - 4x$, so $f'(x) = x^2 - 4$ and $f''(x) = 2x$. For $x^{(t)} \neq 0$, one Newton step gives

$$\begin{aligned}
x^{(t+1)} &= x^{(t)} - \frac{(x^{(t)})^2 - 4}{2x^{(t)}} && \text{(Newton's method with } p = 1) \\
&= \frac{2(x^{(t)})^2 - (x^{(t)})^2 + 4}{2x^{(t)}} && \text{(common denominator)} \\
&= \frac{1}{2} \left(x^{(t)} + \frac{4}{x^{(t)}} \right). && \text{(simplify)}
\end{aligned}$$

From $x^{(0)} = 2.5$:

- $x^{(1)} = \frac{1}{2}(2.5 + 1.6) = 2.05$;
- $x^{(2)} = \frac{1}{2}(2.05 + 1.951219 \dots) = 2.000610$ (to six decimals);
- $x^{(3)} = 2.0000001$ (to seven decimals).

The iterates approach 2, the local minimizer of f (a stationary point, since $f'(2) = 0$, with $f''(2) = 4 > 0$; Theorem 0.16).

Definition 0.15 (Quadratic model). Let f be as in Definition 0.14 and let $\tilde{x}$ be a point, with $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ and $\mathbf{H} \stackrel{\text{def}}{=} \mathbf{H}_f(\tilde{x})$. The **quadratic model** of f at $\tilde{x}$ is the function of a step $\tilde{\delta} \in \mathbb{R}^p$

$$q(\tilde{\delta}) \stackrel{\text{def}}{=} f(\tilde{x}) + \tilde{g}^\top \tilde{\delta} + \frac{1}{2} \tilde{\delta}^\top \mathbf{H} \tilde{\delta},$$

the second-order Taylor polynomial of f at $\tilde{x}$: when the second partial derivatives of f are continuous, Theorem 0.15 says that $f(\tilde{x} + \tilde{\delta})$ equals the same expression with the Hessian evaluated at a point between $\tilde{x}$ and $\tilde{x} + \tilde{\delta}$.

Example 0.50 (The model at $x = 2.5$). For $f(x) = \frac{1}{3}x^3 - 4x$ at $x = 2.5$ (Example 0.49), $g = 2.5^2 - 4 = 2.25$ and $H = [2 \cdot 2.5] = [5]$, so $q(\delta) = f(2.5) + 2.25\delta + 2.5\delta^2$. For $f(\tilde{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, whose Hessian is $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ at every point, the model at any $\tilde{x}$ equals $f(\tilde{x} + \tilde{\delta})$ exactly (Example 0.28 with $\mathbf{S} = \begin{bmatrix} 1/2 & 0 \\ 0 & 3/2 \end{bmatrix}$).

Theorem 0.23 (The Newton step minimizes the quadratic model). *Let q be the quadratic model of f at $\tilde{x}$ (Definition 0.15), with $\mathbf{H} = \mathbf{H}_f(\tilde{x})$ **positive definite**. Its unique minimizer is the Newton step $\tilde{\delta}^* = -\mathbf{H}^{-1}\tilde{g}$.*

i Proof

Proof. $\mathbf{H}$ is symmetric, by the definition of [positive definite](#), and invertible ([positive definite inverse](#)), so $\tilde{\delta}^*$ exists and $\mathbf{H}\tilde{\delta}^* = -\tilde{g}$. For any $\tilde{d} \in \mathbb{R}^p$, first

$$\begin{aligned}\tilde{d}^\top \mathbf{H}\tilde{\delta}^* &= -\tilde{d}^\top \tilde{g} \quad (\mathbf{H}\tilde{\delta}^* = -\tilde{g}) \\ &= -\tilde{g}^\top \tilde{d}, \quad (\text{a } 1 \times 1 \text{ matrix equals its transpose})\end{aligned}$$

and $\tilde{\delta}^{*\top} \mathbf{H}\tilde{d} = \tilde{d}^\top \mathbf{H}\tilde{\delta}^*$ in the same way, using $\mathbf{H}^\top = \mathbf{H}$. Then

$$\begin{aligned}q(\tilde{\delta}^* + \tilde{d}) &= f(\tilde{x}) + \tilde{g}^\top (\tilde{\delta}^* + \tilde{d}) + \frac{1}{2} (\tilde{\delta}^* + \tilde{d})^\top \mathbf{H} (\tilde{\delta}^* + \tilde{d}) && \text{(definition of } q) \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \tilde{g}^\top \tilde{d} + \frac{1}{2} \left(\tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* + \tilde{\delta}^{*\top} \mathbf{H}\tilde{d} + \tilde{d}^\top \mathbf{H}\tilde{\delta}^* + \tilde{d}^\top \mathbf{H}\tilde{d} \right) && \text{(distribute)} \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \tilde{g}^\top \tilde{d} + \frac{1}{2} \tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* - \tilde{g}^\top \tilde{d} + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d} && \text{(both cross terms equal } -\tilde{g}^\top \tilde{d}) \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \frac{1}{2} \tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d} && \text{(cancel } \tilde{g}^\top \tilde{d}) \\ &= q(\tilde{\delta}^*) + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d}, && \text{(definition of } q)\end{aligned}$$

which is greater than $q(\tilde{\delta}^*)$ for every $\tilde{d} \neq \tilde{0}_{p \times 1}$, since $\mathbf{H}$ is positive definite. $\square$

Example 0.51 (The Newton step at $x = 2.5$). For the model $q(\delta) = f(2.5) + 2.25\delta + 2.5\delta^2$ of Example [0.50](#), $H = [5]$ is positive definite, so by Theorem [0.23](#) its minimizer is $\delta^* = -H^{-1}g = -2.25/5 = -0.45$. Setting the derivative $2.25 + 5\delta$ to 0 gives the same δ^* , and $2.5 - 0.45 = 2.05 = x^{(1)}$ (Example [0.49](#)).

Definition 0.16 (Quadratic convergence). A sequence $\tilde{x}^{(0)}, \tilde{x}^{(1)}, \dots$ in $\mathbb{R}^p$ that converges to $\tilde{x}^*$ **converges quadratically** if there is a constant C with

$$\|\tilde{x}^{(t+1)} - \tilde{x}^*\| \leq C \|\tilde{x}^{(t)} - \tilde{x}^*\|^2 \quad \text{for every } t.$$

Example 0.52 (Halving the error is not quadratic convergence). $x^{(t)} = 2^{-t}$ converges to 0, but it does not converge quadratically: the bound would need $2^{-(t+1)} \leq C (2^{-t})^2 = C 2^{-2t}$, that is, $2^{t-1} \leq C$, which fails once $t > 1 + \log_2 C$. Each step only halves the error.

Example 0.53 (The error squares at each step). For the iteration $x^{(t+1)} = \frac{1}{2}(x^{(t)} + 4/x^{(t)})$ of Example [0.49](#),

$$\begin{aligned}
x^{(t+1)} - 2 &= \frac{1}{2} \left(x^{(t)} + \frac{4}{x^{(t)}} \right) - 2 && \text{(the iteration)} \\
&= \frac{(x^{(t)})^2 + 4 - 4x^{(t)}}{2x^{(t)}} && \text{(common denominator)} \\
&= \frac{(x^{(t)} - 2)^2}{2x^{(t)}}. && \text{(the numerator is a perfect square)}
\end{aligned}$$

So near 2 the error is roughly squared and divided by 4 at each step: the errors are 0.05, then 0.00061, then 0.000000093, and $0.05^2/(2 \cdot 2.05) \approx 0.00061$. The iterates converge quadratically (Definition 0.16): $C = \frac{1}{2}$ works for every $x^{(t)} \geq 1$, since then $\frac{1}{2x^{(t)}} \leq \frac{1}{2}$, and every iterate from 2.5 is at least 2, since the identity makes $x^{(t+1)} - 2 \geq 0$ whenever $x^{(t)} > 0$. Zhou (2024) states such a bound for Newton's method in general; it holds near a minimizer where the Hessian is positive definite, for smooth enough f , which these notes do not prove.

Example 0.54 (Newton's method can find a maximum). From $x^{(0)} = -2.5$ the same iteration gives $-2.05, -2.000610, -2.0000001$, approaching -2 . But -2 is not a minimizer of f : it is a stationary point of $-f$, whose second derivative there is $-f''(-2) = 4 > 0$, so it is a strict local minimizer of $-f$ (Theorem 0.16), that is, f is strictly larger at -2 than at all nearby points. Newton's method looks only for a stationary point, and here the Hessian $f''(x) = 2x$ is negative along the way, so the quadratic model has no minimizer (with $H < 0$, $q(\delta) = f + g\delta + \frac{1}{2}H\delta^2$ decreases without bound) and Theorem 0.23 does not apply: the positive definite hypothesis there is needed.

Theorem 0.24 (Newton's method minimizes a quadratic in one step). *Let $\mathbf{S}$ be a $p \times p$ positive definite matrix, $\tilde{c} \in \mathbb{R}^p$, and $f(\tilde{x}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} - \tilde{c}^\top \tilde{x}$. From any $\tilde{x}^{(0)}$, Newton's method (Definition 0.14) gives $\tilde{x}^{(1)} = \mathbf{S}^{-1} \tilde{c}$, the unique minimizer of f .*

i Proof

Proof. Partial derivatives of a sum or of a constant multiple are the sum or multiple of the partial derivatives, so by Theorem 0.7 for $\frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ and Corollary 0.1 for $\tilde{c}^\top \tilde{x}$, $\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) = \frac{1}{2} (2\mathbf{S} \tilde{x}) - \tilde{c} = \mathbf{S} \tilde{x} - \tilde{c}$; by Theorem 0.11 with $\frac{1}{2} \mathbf{S}$ in place of $\mathbf{S}$, and because the gradient $-\tilde{c}$ of the linear term is constant (Definition 0.4), $\mathbf{H}_f(\tilde{x}) = \mathbf{S}$, which is invertible (positive definite inverse). So

$$\begin{aligned}
\tilde{x}^{(1)} &= \tilde{x}^{(0)} - \mathbf{S}^{-1}(\mathbf{S}\tilde{x}^{(0)} - \tilde{c}) && \text{(definition of Newton's method)} \\
&= \tilde{x}^{(0)} - \mathbf{S}^{-1}\mathbf{S}\tilde{x}^{(0)} + \mathbf{S}^{-1}\tilde{c} && \text{(distribute)} \\
&= \tilde{x}^{(0)} - \tilde{x}^{(0)} + \mathbf{S}^{-1}\tilde{c} && (\mathbf{S}^{-1}\mathbf{S} = \mathbf{I}_p) \\
&= \mathbf{S}^{-1}\tilde{c}. && \text{(cancel)}
\end{aligned}$$

The gradient at $\mathbf{S}^{-1}\tilde{c}$ is $\mathbf{S}\mathbf{S}^{-1}\tilde{c} - \tilde{c} = \tilde{0}_{p \times 1}$, and f is strictly convex (Theorem 0.21, part 2), so $\mathbf{S}^{-1}\tilde{c}$ is a global minimizer (Corollary 0.4). It is the only one: a strictly convex function has at most one global minimizer, since the midpoint of two distinct global minimizers, which have equal values, would have a strictly smaller value (Definition 0.11 with $t = \frac{1}{2}$). $\square$

Example 0.55 (One step to the minimizer). Let $f(\tilde{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, so $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ and $\tilde{c} = \tilde{0}_{2 \times 1}$. From $\tilde{x}^{(0)} = (3, 1)^\top$ the gradient is $\mathbf{S}\tilde{x}^{(0)} = (3, 3)^\top$, and $\mathbf{S}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}$, so the Newton step is $-\mathbf{S}^{-1}(3, 3)^\top = -(3, 1)^\top$ and $\tilde{x}^{(1)} = (3, 1)^\top - (3, 1)^\top = \tilde{0}_{2 \times 1}$, the minimizer.

Definition 0.17 (Gradient descent). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives, let $\tilde{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \dots$ be positive **step lengths**. **Gradient descent** computes, for $t = 0, 1, 2, \dots$,

$$\tilde{x}^{(t+1)} \stackrel{\text{def}}{=} \tilde{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}).$$

It is Newton's method (Definition 0.14) with the Hessian replaced by $\frac{1}{s^{(t)}}\mathbf{I}_p$, so it needs no second derivatives. Choosing $s^{(t)}$ to minimize $f(\tilde{x}^{(t)} - s \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}))$ over $s > 0$ is **exact line search**.

Example 0.56 (Fixed steps on a parabola). For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length s is $x^{(t+1)} = x^{(t)} - 2s x^{(t)} = (1 - 2s)x^{(t)}$, and $x^{(t)} = (1 - 2s)^t x^{(0)}$.

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Theorem 0.25 (A short enough step downhill decreases the function). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (Definition 0.7), and let $\tilde{x} \in \mathbb{R}^p$ be a point with $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}) \neq \tilde{0}_{p \times 1}$. Then there is an $\bar{s} > 0$ with $f(\tilde{x} - s\tilde{g}) < f(\tilde{x})$ for every $s \in (0, \bar{s})$.*

i Proof

Proof. Let $h(s) \stackrel{\text{def}}{=} f(\tilde{x} - s\tilde{g})$. The inner function $s \mapsto \tilde{x} - s\tilde{g}$ has derivative $-\tilde{g}^\top$ (Definition 0.3), so by the vector chain rule (Theorem 0.8)

$$\begin{aligned} h'(s) &= -\tilde{g}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{x} - s\tilde{g}), & (\text{chain rule}) \\ h'(0) &= -\tilde{g}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{x}) & (\text{set } s = 0) \\ &= -\tilde{g}^\top \tilde{g} & (\text{definition of } \tilde{g}) \\ &= -\|\tilde{g}\|^2 & (\text{squared length}) \\ &< 0. & (\tilde{g} \neq \tilde{0}_{p \times 1}) \end{aligned}$$

Since $h'(0) = \lim_{s \rightarrow 0} (h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\tilde{x} - s\tilde{g}) < f(\tilde{x})$. $\square$

Example 0.57 (How short is short enough). In Example 0.56, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1 - 2s)^2 x^2 < x^2$ exactly when $|1 - 2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.25 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

Example 0.58 (Gradient descent zigzags). Let $f(\tilde{x}) = \frac{1}{2}(x_1^2 + bx_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) = (x_1, bx_2)^\top$ and one gradient step of length s from $\tilde{x}$ gives $((1-s)x_1, (1-bs)x_2)^\top$. For exact line search, let $\phi(s) \stackrel{\text{def}}{=} f$ at that point:

$$\begin{aligned} \phi(s) &= \frac{1}{2}((1-s)^2 x_1^2 + b(1-bs)^2 x_2^2), & (\text{definition of } f) \\ \phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 & (\text{chain rule, term by term}) \\ &= -x_1^2 + sx_1^2 - b^2x_2^2 + b^3sx_2^2 & (\text{expand the products}) \\ &= s(x_1^2 + b^3x_2^2) - (x_1^2 + b^2x_2^2). & (\text{collect the terms in } s) \end{aligned}$$

For $\tilde{x} \neq \tilde{0}_{2 \times 1}$, $x_1^2 + b^3x_2^2 > 0$, so ϕ' is increasing in s with a single zero, and ϕ is minimized at $s^* = \frac{x_1^2 + b^2x_2^2}{x_1^2 + b^3x_2^2}$. From $\tilde{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned} s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} \quad (\text{substitute } x_1 = b, x_2 = 1) \\ &= \frac{2}{1+b}, \quad (\text{cancel } b^2) \end{aligned}$$

and, with $r \stackrel{\text{def}}{=} \frac{b-1}{b+1}$,

$$\begin{aligned} 1 - s^{(0)} &= \frac{1+b-2}{1+b} = r, & (\text{common denominator}) \\ 1 - b s^{(0)} &= \frac{1+b-2b}{1+b} = -r, & (\text{common denominator}) \\ \tilde{x}^{(1)} &= (r b, -r \cdot 1)^\top = r(b, -1)^\top. & (\text{the gradient step from } (b, 1)^\top) \end{aligned}$$

For $b \neq 1$ the same holds at every step, by induction (when $b = 1$, $r = 0$ and $\tilde{x}^{(1)}$ is already the minimizer): if $\tilde{x}^{(t)} = (c b, \pm c)^\top$ for a number $c \neq 0$, then s^* , which depends only on $x_1^2 = c^2 b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor c^2 cancels), and the step multiplies the first entry by r and the second by $-r$. So

$$\tilde{x}^{(t)} = (b r^t, (-r)^t)^\top, \quad f(\tilde{x}^{(t)}) = \frac{1}{2}(b^2 + b) r^{2t} = r^{2t} f(\tilde{x}^{(0)}).$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\tilde{0}_{2 \times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\tilde{x}^{(1)} = (1.5, -0.5)^\top$ and $\tilde{x}^{(2)} = (0.75, 0.25)^\top$, and f falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so f shrinks by only about 4% per step. Newton's method reaches the minimizer of any such f in one step (Theorem 0.24).

Further reading

- Marsden and Tromba (2013) is a standard textbook on multivariable and vector calculus. It covers differentiation of functions of several variables, multiple integrals, line and surface integrals, and the theorems of Green, Gauss, and Stokes.

See also the [Linear Algebra and Vector Calculus further reading](#).

- [Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" \(2023 Fall\)](#)

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